

Tidal interactions in star cluster simulations

Rosemary A. Mardling^{1★} and Sverre J. Aarseth²

¹*Mathematics Department, Monash University, Clayton 3800, Victoria, Australia*

²*Institute of Astronomy, University of Cambridge, Cambridge*

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ABSTRACT

We describe the implementation of tidal circularization of binaries in an N -body code for star cluster simulations. The first part contains the theoretical framework for normal and chaotic tidal interactions, including capture from hyperbolic orbits. This formulation yields convenient expressions which are used to modify the binary elements. Stars are represented as polytropes, with a time-dependent effective polytropic index calculated for evolving stars. Stellar evolution is treated using a fast look-up table for stellar masses and radii. This gives a consistent astrophysical description of open clusters containing a significant proportion of primordial binaries with a wide range of masses and periods. An analytic expression for the chaos boundary for arbitrary mass ratio and polytropic indices is presented. We provide detailed correction procedures for tidal circularization and chaotic motion for perturbed binaries which are studied by the classical Kustaanheimo–Stiefel two-body regularization method and also outline a similar treatment for multiple regularization of temporary subsystems involving 3–6 members. Strong interactions in the latter lead to the formation of chaotic binaries and stable hierarchical systems in which the eccentricity of the inner binary may be subject to systematic changes on relatively short time-scales. Finally, we illustrate the effect of tidal circularization by presenting some results of a realistic cluster simulation involving 10^4 single stars and 500 primordial binaries.

Key words: gravitation – stellar dynamics – binaries: close – stars: evolution – stars: mass-loss – open clusters and associations: general.

1 INTRODUCTION

The dynamics in a star cluster depends vitally on the binary fraction in its core, and equally important is the distribution of semimajor axes. The cross-section that a binary presents to passing stars and other binaries determines the strength of the interaction as well as its character. While the majority of interactions give rise to small perturbations whose cumulative effect governs the overall evolution of the cluster core, some interactions may involve exchange, collision or perhaps form a hierarchical system. Strong interactions are extremely important for understanding the diversity of exotic objects found in globular cluster cores, in particular millisecond pulsars, low-mass X-ray binaries, cataclysmic variables and blue stragglers. They may also be responsible for some of these objects in the field; it is possible that they were ejected from old open clusters following strong interactions (Mardling 1996a). It is therefore important to model the *evolution* of the semimajor axes as accurately as possible in order to delineate the various processes. The semimajor axes of binaries close enough to interact tidally generally shrink so that they become less effective as heat sources for the core, while at the

same time mass segregation tends to increase the binary fraction in the central regions.

Tidal interaction is important for three broad categories of binaries. Some binaries may undergo complete circularization during the pre-main sequence stage (a few per cent in our sample). We apply the circularization procedure to the initial population of binaries to obtain a reasonable initial distribution of eccentricities and orbital periods. While some binaries continue to interact tidally on the main sequence, most binaries which do this at some stage will do so when one star evolves up the giant branch or asymptotic giant branch. The third category of binaries whose orbits may evolve tidally and in particular, may interact chaotically, are those which interact strongly with other stars or star systems. Such interactions include the fly-by or resonant interaction of a third star, with the latter possibly involving exchange; the collision of two binaries, or indeed a binary and a stable triple system. We may also include tidal capture in this category.

A fourth category may be added to this list: binaries which survive neutron star formation. While most binaries in which a neutron star is born dissociate in response to the velocity kick applied at birth, those which survive are, in the majority of cases and by necessity, highly eccentric and extremely close at periastron (Brandt & Podsiadlowski 1995). Thus we may expect many of these stars to interact chaotically.

★ E-mail: mardling@monash.edu.au

The present paper develops the necessary framework for the implementation of tidal circularization. This formulation divides into two parts. The first part is for normal tidal evolution, and is based on the formulation given by Hut (1981) for arbitrary orbital eccentricities (but here we neglect stellar spin). The second part is for chaotic tidal evolution, and is based on the theory presented in Mardling (1995a,b). It suggests that for a range of binaries which are eccentric and very close at periastron, energy exchange between the orbit and the tides is chaotic. In particular and in contrast to normal tidal interaction, the tidal energy may build up to the stage where it is a substantial fraction of the binding energy of the star. However, if the binary is near the chaos boundary when it becomes chaotic, the tidal energy need not build up to much more than for normal tidal interactions. This is particularly important for binaries formed by tidal capture. It is still not clear how stars in chaotic orbits will respond to the huge tidal energies implied, if indeed the tides are not damped entirely via non-linear interactions between the modes of oscillation (Kumar & Goodman 1996), although it is hard to see how such behaviour can occur without a large build-up of tidal energy in the first place. It is generally agreed that the response of the stars, including expansion and mass loss, depends on where the tidal energy is deposited (Podsiadlowski 1996).

The chaos theory of tidal interaction was developed for the study of binary formation via tidal capture, a process believed to be important for forming close binaries in globular clusters. The core stellar densities in such clusters are such that one expects of the order of tens to hundreds of captures to occur per cluster (Davies & Benz 1995). Many of these will result in collisions, while some may survive the circularization process, depending on how the stars respond to the process. Some of these ‘successful’ captures will involve neutron stars and may be an important formation mechanism for the low-mass X-ray binaries seen in relative abundance in globular clusters (see next paragraph). It has been suggested (Mardling 1996a) that *old open* clusters may also be important sites for tidal capture to occur, even though the core densities are considerably lower than globular cluster core densities. Although capture is very much less likely to occur, the large numbers of open clusters both now and in the past, together with their lower velocity dispersions, may mean that they provide nurseries for some exotic objects. For those objects containing neutron stars, the biggest problem with this formation scenario is the retention of newly born neutron stars by these clusters. However, one may appeal to the fact that globular clusters retain many neutron stars in spite of the fact that the escape velocities of these clusters are much lower than most kick velocities derived from pulsar statistics (see, for example, Hansen & Phinney 1997).

With typical memberships of up to 10^4 stars, open clusters can be studied by direct N -body methods without recourse to any scaling. It is therefore appropriate to begin our application of the circularization theory with these clusters which are so well suited to this purpose.

The plan of our paper is as follows. Section 2 contains the main theoretical development of the implementation of tidal circularization, both normal and chaotic. In Section 3 we describe a variety of technical procedures for implementation of these expressions in an N -body code. Particular emphasis is placed on the regularized two-body formulation which is ideally suited to the present approach. It is fortuitous that a synthetic scheme for stellar evolution is already in place (Tout et al. 1997), thus permitting a consistent astrophysical treatment. We find that hierarchical systems play an important role in the cluster evolution,

and various aspects connected with such configurations are discussed in Section 4. Some recent results are illustrated in Section 5, and Section 6 provides a general discussion.

2 TIDAL INTERACTIONS

We distinguish between two types of tidal interactions. *Normal* tidal interactions generally involve small tides ($\delta R/R < 1$ per cent, say), except for systems close to Roche contact. Energy and angular momentum are exchanged between the orbit and the stars in a quasi-periodic fashion, generally on the time-scale of a few orbital periods. Such exchanges are purely dynamical, while secular exchanges owing to energy loss from the system occur on time-scales many orders of magnitude longer. The tides induced in a star by a binary companion can be thought of as being composed of an *equilibrium tide* and a *dynamical tide*. The equilibrium tide corresponds to instantaneous hydrostatic equilibrium, while the dynamical tide allows for resonances between the natural frequencies of the stellar modes and the tidal frequency. The word ‘equilibrium’ is generally a misnomer since no equilibrium configurations exist other than for circular binaries with synchronously rotating components.

Two different dissipation mechanisms have been identified as important (Zahn 1975, 1989). Stars with radiative envelopes (early type stars) dissipate tidal energy via radiative damping of the dynamical tide, while a much more efficient process exists for stars with convective envelopes, i.e. low-mass main-sequence stars and giants. If the convective turnover time is comparable to the time of periastron passage,¹ then coupling can occur between the equilibrium tide and convective motions and result in damping via a turbulent viscosity.

It is assumed in these models that the dynamical tide is generally much less energetic than the equilibrium tide, this being based on calculating the response of the star after one periastron passage. One of us (RM) has used a self-consistent model (Gingold & Monaghan 1980; Mardling 1995a) to show that it is possible for the dynamical tide to be at least as energetic as the equilibrium tide. The model is self-consistent in the sense that the orbit and tides are free to exchange energy in either direction. This results in energy exchange at periastron depending on the tidal energy already present, as well as the phase of oscillation. The most extreme consequence of this is the existence of a *chaotic* interaction between the tides and the orbit.

Any kind of damping will cause the tidal bulge to lag (or lead) with respect to the line joining the centres of the two stars. The resulting torque allows transfer of angular momentum between the orbit and the stars; the direction will be such that the stars tend towards having spins which are synchronous with the orbital velocity at periastron. A simple and elegant model exists which describes the evolution of the orbit and stellar spin via the equilibrium tide for arbitrary eccentricity (Hut 1981). The physics of the damping process is not addressed, but rather is treated phenomenologically via a damping time-scale to be prescribed by the user of the model. To this end, we characterize this time-scale via a star’s Q value² (see Section 2.1.1). Since reliable estimates for main sequence stars are not available (Gough 1997) we take $Q = 10^4$ for main sequence stars; probably too low for late-type main sequence stars, and too high for those of early type.

¹ $2\pi/\Omega_p$, with Ω_p being the orbital frequency at periastron.

² Defined to be the e-folding number of oscillations needed to damp the tidal energy. See Goldreich & Soter (1966).

While the Hut model is not designed to describe orbital evolution as a result of dissipation of the dynamical tide, it is none the less adopted to describe systems for which this is the dominant damping mechanism. This choice is justified for our purposes because (a) the Hut model is mathematically simple to implement in this preliminary study and there exists no equivalent model for the dynamical tide; and (b) although the damping mechanisms are different, they both depend on the strength of the forcing which is implicit in the Hut model. Since most of the uncertainty lies in the choice of Q values for the stars, this seems a reasonable first approximation.

While transfer of angular momentum from the orbit to the stellar spin will result in a closer final configuration, this effect is small for most systems. To see this consider the following. Let J_i and J_f be the initial and final total angular momenta of a binary (assuming that any stellar spin is aligned with the orbit axis). If no angular momentum is transferred to stellar spin, then (assuming as well that there is no initial spin and that the binary circularizes completely)

$$J_i = \mu[Gm_{12}R_p^i(1+e)]^{1/2} = J_f = \mu[Gm_{12}a_f]^{1/2}, \quad (1)$$

where m_1 and m_2 are the stellar masses, μ is the reduced mass, $m_{12} = m_1 + m_2$, e is the initial eccentricity, R_p^i is the initial periastron separation and a_f is the final semimajor axis. Thus in the case of no stellar spin,

$$\frac{a_f}{R_p^i} = 1 + e. \quad (2)$$

If some angular momentum is transferred to the stellar spin of one star (say, star 1)

$$J_f = \mu[Gm_{12}a_f]^{1/2} + I\Omega, \quad (3)$$

where I is the moment of inertia of star 1 and Ω is its final spin rate. For synchronous spin,

$$\Omega = \sqrt{\frac{Gm_{12}}{a_f^3}} = \sqrt{\frac{Gm_1}{r_1^3}} \sqrt{\frac{1+q}{(a_f/r_1)^3}}, \quad (4)$$

where r_1 is the final radius of star 1 and $q = m_2/m_1$ is the mass ratio. If in addition star 1 fills its Roche lobe, then (Eggleton 1983)

$$\frac{r_1}{a_f} = \frac{0.49}{0.6 + q^{2/3} \ln(1 + q^{-1/3})} \equiv f(q), \quad (5)$$

so that

$$J_f = m_1 \sqrt{Gm_{12}a_f} \left[\frac{q}{1+q} + \alpha f(q)^2 \right], \quad (6)$$

where α is defined such that $I = \alpha m_1 r_1^2$ ($\alpha = 0.4$ for a homogeneous body). Thus

$$\frac{a_f}{R_p^i} = \phi(1+e), \quad (7)$$

where ϕ is the ‘diminishment factor’ given by

$$\phi = [1 + \alpha f(q)^2 (1+q)/q]^{-2}. \quad (8)$$

For a giant of mass $1.3 M_\odot$, $\alpha = 0.09$ while the same star on the asymptotic giant branch has $\alpha = 0.07$ (Verbunt, Mardling & Frost 1997). In these cases, $\phi = 0.95$ and 0.96 respectively for $q = 1$, while for the extreme mass ratio $q = 0.1$, $\phi = 0.56$ and 0.63 . In the other extreme, $\phi \rightarrow 1$ as q becomes large. Although most binaries will have $q < 1$, binaries containing stellar remnants may

have $q > 1$. Cool giants can suffer a density inversion at the end of their evolution and may thus have values for α closer to 0.4 (Podsiadlowski, private communication). However, such stars tend to shrink during this phase so that if they ever fill their Roche lobes, they would have done so before this phase. Such low-mass giants are found in globular clusters, which we hope to be able to model in the near future. Fig. 1 plots the diminishment factor ϕ against q for the various values of α for systems which fill their Roche lobes. Systems which do not fill their Roche lobes will have values for α closer to unity. Thus neglecting stellar spin may lead us to underestimate the number of systems which experience Roche-lobe overflow; however, the effect on the cluster dynamics is likely to be minimal.

The second kind of tidal interaction we consider involves chaotic orbital behaviour. Energy exchanges between the orbit and the stars are unpredictable, allowing the tides to become extremely energetic so that the orbit can partially circularize dynamically. Permanent circularization is only possible via dissipation, and since the tides can be very energetic, non-linear effects are taken into account via a non-linear damping time-scale which is likely to be very much shorter than the linear damping time-scale.

Chaotic behaviour generally occurs for binaries which are highly eccentric and extremely close at periastron. Fig. 2 shows the boundary between chaotic and normal orbits for equal-mass

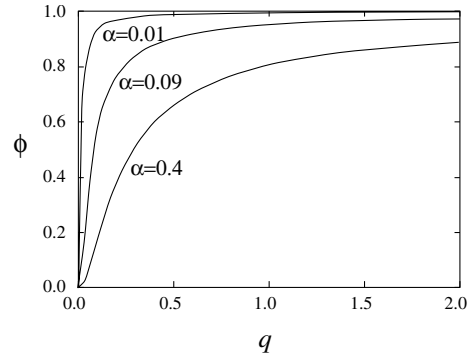


Figure 1. Diminishment factor ϕ for a range of mass ratios and moments of inertia for systems which fill their Roche lobes.

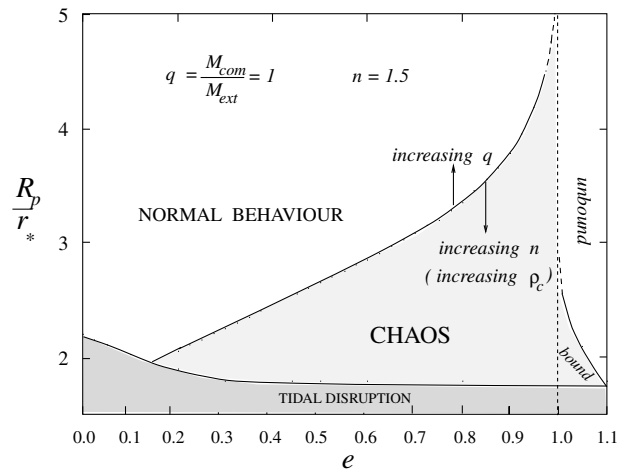


Figure 2. The chaos boundary for equal mass binaries consisting of a point mass and a polytrope of index 1.5. A point in this diagram represents the initial binary configuration. The hyperbolic branch corresponds to tidal captures.

binaries modelled by a point mass and a polytrope of index 1.5. The boundary moves down for more centrally condensed stars and for binaries with mass ratios (point-mass to extended star) less than unity.

A chaotic binary will eventually dissipate sufficient energy to cross the chaos boundary, after which it will slowly circularize normally. Fig. 3 illustrates how a binary evolves through a chaotic phase followed by a normal circularization phase. This case is for an equal-mass binary with an initial eccentricity of unity and an initial periastron separation of three stellar radii. Both stars are polytropes of index 1.5. This plot was produced using the procedures described in this paper.

2.1 Normal tidal circularization

In this section, we outline the procedure for implementing normal (non-chaotic) tidal circularization.

Consider a binary consisting of stars with masses and radii m_1 , m_2 , r_1 and r_2 with $r_1 \geq r_2$, whose structure is represented by polytropes of indices n_1 and n_2 . The state of the binary at time t is given by the periastron separation R_p and the eccentricity e which we represent by (p, e) , where $p = R_p/r_1$. Stellar spin is ignored in our model and hence the orbital angular momentum of an unperturbed binary remains constant during normal tidal circularization. Thus assuming there is no mass or angular momentum loss from the system, the present formulation is such that the state at time t_0 defines all future states. The advantage of this is that it is not necessary to integrate the orbit directly so that (non-chaotic) tidally interacting binaries are nearly as inexpensive computationally as unperturbed tidally inert binaries. If a binary is perturbed, its angular momentum is changed and its future evolution must then be referred to its state at the time of the perturbation (see Section 3.4). In the following, we formulate a procedure which allows us to calculate e and R_p directly at time t , given their values at time t_0 (assuming unperturbed motion).

The orbital angular momentum is given by

$$J_{\text{orb}} = \left(\frac{G m_1^2 m_2^2}{m_1 + m_2} R_p (1 + e) \right)^{1/2} \quad (9)$$

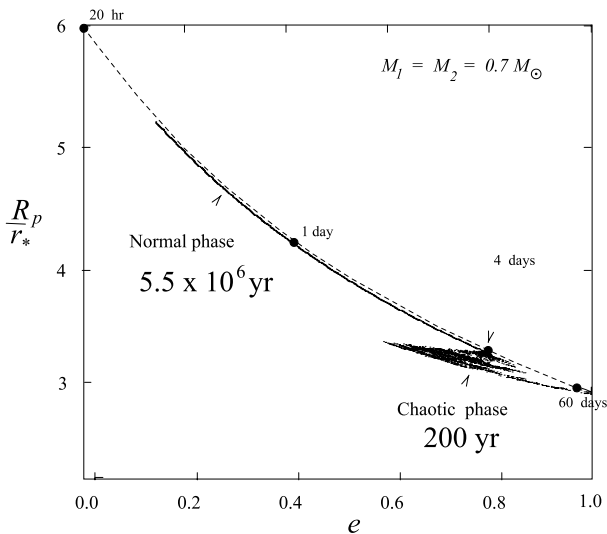


Figure 3. The tidal evolution of a chaotic binary. After a short chaotic phase the binary enters a long period of normal tidal circularization. This plot was produced using the procedures described in this paper.

which remains constant so that

$$p(1 + e) = p_0(1 + e_0), \quad (10)$$

where p_0 and e_0 are the scaled periastron separation and eccentricity at time t_0 . Assuming that the equilibrium tide governs the evolution, the rate of change of eccentricity is given by (see Hut 1981, equation A15)³

$$\frac{de}{dt} = -\frac{9}{2} \left[\frac{k_1}{T_1} q(1+q) \left(\frac{r_1}{a} \right)^8 + \frac{k_2}{T_2} \frac{1}{q} \left(1 + \frac{1}{q} \right) \left(\frac{r_2}{a} \right)^8 \right] \times \frac{e}{(1 - e^2)^{13/2}} \left(1 + \frac{15}{4} e^2 + \frac{15}{8} e^4 + \frac{5}{64} e^6 \right), \quad (11)$$

where $q = m_2/m_1$, a is the semimajor axis of the orbit, k_1 and k_2 are the apsidal motion constants of stars 1 and 2 respectively, and T_1 and T_2 are their damping time-scales defined in terms of tidal lag angles. These last four quantities are properties of the stars, and since we are using polytropes to model stars, we need to express them in terms of structure constants associated with polytropes. The structure constants we use are the oscillation frequencies of the $l = 2$ f -modes, w_1 and w_2 for star 1 and star 2 respectively and the associated overlap integrals (Press & Teukolsky 1977) Q_1 and Q_2 , which are integrals over the polytrope of its normal modes. Table 1 lists the constants we use here and in the next section, with w_s^2 in units of Gm_*/r_*^3 and Q_s in units of $m_* r_*^2$, for polytropes of index 1.5 and 3 (Lee & Ostriker 1986). Here, r_* and m_* are the radius and mass of the star in question. One can show (Schwarzschild 1965; Kumar, Ao & Quartaert 1995) that

$$k_i = \frac{2\pi}{5} \frac{G}{r_i^5} \frac{Q_i^2}{w_i^2} = \frac{2\pi}{5} \frac{Q_{s,i}^2}{w_{s,i}^2}, \quad i = 1, 2. \quad (12)$$

2.1.1 Damping time-scales

The damping time-scales, T_i , used in the Hut model (also referred to as the weak friction model) are defined in terms of the tidal lag times, τ_i (the time it takes for the tidal bulge to catch up with the current position of the line of centres), which in turn are defined in terms of the tidal lag angles and hence in terms of the damping constants of the stars. In particular, the lag time is taken as constant over the evolution. For star i , we have

$$\frac{T_*}{T_i} = \frac{\tau_i}{T_*}, \quad (13)$$

where $T_* = \sqrt{r_*^3/Gm_*}$. Thus, the less effective the damping, the smaller the lag time and hence, the longer the damping time-scale. The response of a forced damped oscillator will lag behind the forcing by an angle

$$\phi = -\frac{2\mathcal{C}\Omega}{\omega^2}, \quad (14)$$

where $\mathcal{C}$ is the damping constant of the oscillator, ω is its natural frequency and Ω is the forcing frequency, which we have assumed to be significantly less than the oscillator frequency. For non-rotating stars, the lag angles ϕ_i are given by (see Fig. 4)

$$\phi_i = -\Omega\tau_i. \quad (15)$$

As a matter of convenience, we choose to define the damping

³ Since we are assuming that no angular momentum is transferred to the stellar spin, only the radial force will be active in circularizing the orbit.

Table 1. Structure constants for polytropes (f -modes).

n	l	w_s^2	Q_s
1.5	2	2.119	0.4909
	3	3.742	0.4677
3	2	8.175	0.2372
	3	9.413	0.1519

time-scales in terms of the $l = 2$ f -mode and some nominal Q value. For main-sequence stars we choose $Q = 10^4$, while for giants, Q is calculated in terms of the luminosity (see Section 2.6). In effect, we are assigning all the uncertainty of the physics of tidal damping to these Q values. Equating equations (14) and (15), we obtain

$$\frac{2C_i\Omega}{\omega_i^2} = \Omega\tau_i = \frac{T_*^2}{T_i}\Omega, \quad (16)$$

so that

$$\frac{1}{T_i} = \frac{2C_i}{\omega_{s,i}^2}, \quad (17)$$

where $\omega_{s,i}$ is the scaled oscillation frequency for star i and $C_i = \omega_i/2\pi Q_i$.

Using $a/r_1 = p/(1 - e)$ and equation (10), we can now write the eccentricity evolution equation as

$$\begin{aligned} \frac{de}{dt} = & -2 \left[C_1 \frac{Q_{s,1}^2}{\omega_{s,1}^4} q(1+q) + C_2 \frac{Q_{s,2}^2}{\omega_{s,2}^4} \frac{1}{q} \left(1 + \frac{1}{q}\right) \left(\frac{r_2}{r_1}\right)^8 \right] \\ & \times \frac{f(e)}{p_0^8(1+e_0)^8} \equiv -C_{\text{eff}}f(e), \end{aligned} \quad (18)$$

where

$$f(e) = \frac{9\pi}{5} e(1 - e^2)^{3/2} \left(1 + \frac{15}{4}e^2 + \frac{15}{8}e^4 + \frac{5}{64}e^6 \right). \quad (19)$$

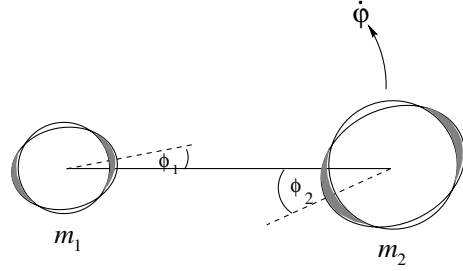
The solution to this equation is given formally by

$$t - t_0 = -\frac{1}{C_{\text{eff}}} \int_{e_0}^e \frac{de}{f(e)} = \frac{1}{C_{\text{eff}}} (I(e) - I(e_0)), \quad (20)$$

where C_{eff} is defined in equation (18) and the function I satisfies

$$\frac{dI}{de} = -\frac{1}{f(e)}. \quad (21)$$

Our aim is to invert equation (20) for e as a function of t which we do in three steps as follows. First, we choose a solution from the family of solutions which satisfy equation (21) and which covers a large enough range of e for our purposes. Second, we fit this solution with a quadratic rational function. Finally, this can be inverted algebraically so that given e_0 and p_0 at t_0 , the current value of e can be found. Two solutions of equation (21) are plotted in Fig. 5 for the initial conditions $(e, I) = (0.939, 0)$ and $(e, I) = (0.995, 0)$. We interpret these solutions as follows, recalling that a change in I is proportional to a change in t . For extremely high eccentricities, at least in the equilibrium tide approximation, a binary only damps effectively near periastron. Thus the circularization rate is slow for such eccentricities, as can be seen for the solution with $e_0 = 0.995$. Such a solution is hard to fit with a quadratic rational function, this choice of fitting function being made for its relative accuracy at the same time as being invertible

**Figure 4.** Tidal lag. The tidal bulges of the stars lag behind the line of centres by an amount which is inversely proportional to their tidal damping time-scale.

algebraically. Since most binaries in our cluster model with such high eccentricities are chaotic and hence will involve a different circularization procedure, we choose to sacrifice accuracy at this high eccentricity end and select a solution of equation (21) which models all other values of e well. We choose a solution whose rational function approximation has the value zero at $e = 1$ and this corresponds to choosing $e_0 = 0.939$. At the low eccentricity end, circularization is slow. In fact, complete circularization is approached asymptotically, as can be seen from equations (18) and (19) for small e . With this in mind (and also for technical reasons), we need to choose an eccentricity for which we declare the binary to have circularized. This we take as $e = 0.002$, where the numerical solution gives $I(0.002) = 0.99$. Hence we choose our fitting function, $\tilde{I}(e)$, to be such that $\tilde{I}(0) = 1$.

The quadratic rational function is of the form

$$\tilde{I}(e) = \frac{1 + a_1e + a_2e^2}{1 + b_1e + b_2e^2}, \quad (22)$$

where the coefficients a_i and b_i are given in Table B1 of Appendix B. These values were obtained by choosing four points from the numerical solution which gave the best fit. Putting $\tilde{I}(e) = x$, the inverse function is given by

$$e = \tilde{I}^{-1}(x) = \frac{-1 + c_1x - \sqrt{c_2x^2 + c_3x + c_4}}{c_5 + c_6x}, \quad (23)$$

where the values of c_i are also given in Table B1 of Appendix B.

Thus given the initial state of a binary, (p_0, e_0) at time t_0 , as well as the current time t , we can use equations (20), (22) and (23) to calculate the current eccentricity:

$$e = \tilde{I}^{-1}\{C_{\text{eff}}(t - t_0) + \tilde{I}(e_0)\}. \quad (24)$$

The current periastron separation is then given by equation (10). This simple prescription eliminates the need for integration of the binary orbits, at least when they are unperturbed.

2.2 Circularization times

We can use equation (20) with $e = 0$ to estimate the current circularization time, t_{circ} , for a binary. From equation (22) we have $\tilde{I}(0) = 1$ so that from equation (20),

$$t_{\text{circ}} = \frac{1}{C_{\text{eff}}}(1 - \tilde{I}(e_0)). \quad (25)$$

This will change if the binary has an interaction, or loses mass, or one of the stars evolves so that the periastron separation in terms of the stellar radius decreases. In this case, t_{circ} reduces rapidly

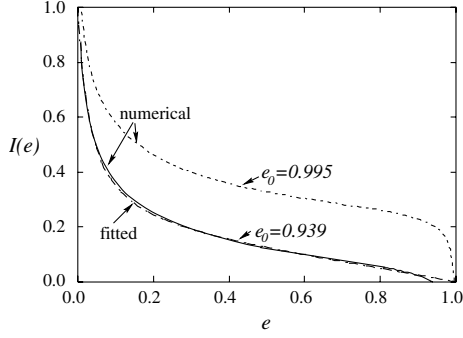


Figure 5. The function $I(e)$. See text for discussion.

because of the steep dependence of C_{eff} on p_0 . Tidal circularization is activated if the estimated circularization time is less than 2 Gyr.

2.3 The chaos boundary

The boundary between chaotic and regular orbital behaviour may be described using a single dimensionless variable ξ which is similar to the generalized Press & Teukolsky similarity variable ζ described in Appendix A. Given the mass ratio, the periastron separation, the eccentricity and the stellar type, we find that the chaos boundary is well described by

$$\frac{a}{r_*} = \alpha \xi^\beta, \quad (26)$$

where $\alpha = 0.048$ and $\beta = 4.0$ are empirically determined constants. Here a is the semimajor axis and r_* is the radius of the ‘dominant’ star which is not necessarily the star with the largest radius (see next paragraph). ξ is the (modified) ratio of the angular velocity at periastron to the frequency of oscillation, ω_* , of the most energetic mode of the dominant star (taken here to be the $l = 2$ f -mode), i.e.

$$\xi = \left[\frac{\omega_s^2}{(1+q)} \frac{p^3}{(1+e)^{1.2}} \right]^{1/2}, \quad (27)$$

where q is the mass ratio of the secondary to the ‘dominant’ star, $w_s = w_*/\sqrt{Gm_*/r_*^3}$, m_* is the mass of the dominant star and $p = R_p/r_*$. The power 1.2 in this expression plays a similar role to $\alpha(\eta)$ (equation A5) in Appendix A. Equations (26) and (27) together give an equation for the chaos boundary:

$$p = C \left[\frac{(1+e)^{1.2}}{(1-e)^{1/2}} \frac{(1+q)}{\omega_s^2} \right]^{2/5}, \quad (28)$$

where $C = 1.836$, and we are assuming $\beta = 4$ exactly. This curve is plotted in Fig. 6 for mass ratios $q = 1$ and $q = 2$ and for polytropes of index 1.5 and 3, which compares these with empirical data obtained using the model described in Mardling (1995a). Note that in reality the chaos boundary is not well defined, but rather has a fractal-like structure. The fit is very good for $n = 1.5$ but only reasonable for $n = 3$. None the less, it is convenient to use this simple formulation for the chaos boundary.

It is found empirically (Mardling 1991) that the chaos boundary for a point mass-polytrope system is not much affected if one replaces the point mass with another polytrope identical to the first. Thus, replacing the point mass with a polytrope of the same index but with any radius smaller than the first polytrope will similarly not affect the chaos boundary much. To use equation

(26), one must decide which star is dominant in the binary, that is, which star is capable of absorbing the most tidal energy; this star determines the chaos boundary. For stars with the same polytropic index, the star with the largest radius dominates, but when the stellar types differ, it is possible for the smaller radius star to dominate. Using the dominant star in equation (26) will yield chaos for a wider range of binary parameters than using its companion, i.e. the chaos boundary in the $(R_p - e)$ plane will be higher (see Fig. 2). To decide which star is dominant, we use the fact that the quantity $(a/r_*)\xi^{-\beta}$ is invariant for any system. From equation (28) we have, for any value of e ,

$$p_1 \left[\frac{1+q}{\omega_{s,1}^2} \right]^{-2/5} = p_2 \left[\frac{1+1/q}{\omega_{s,2}^2} \right]^{-2/5}, \quad (29)$$

where $q = m_2/m_1$, p_1 and $\omega_{s,1}$ are p and ω_s in equation (28) scaled to star 1, while p_2 and $\omega_{s,2}$ are scaled to star 2. In order to compare chaos boundaries, we need to scale the periastron separation using the same star. Using star 1 to scale and writing $p'_2 = (r_2/r_1)p_2$, we have

$$\frac{p_1}{p'_2} = \left(q \frac{\omega_{s,2}^2}{\omega_{s,1}^2} \right)^{2/5} \frac{r_1}{r_2}. \quad (30)$$

Thus if $p_1/p'_2 > 1$, star 1 dominates and is used for the chaos boundary, otherwise star 2 is used.

2.4 Chaotic orbits

For orbits which become chaotic, it is necessary to follow the evolution in detail. It is possible to mimic the chaotic dynamics of a binary using a mapping which gives the energy transfer to, or from, the modes of oscillation as well as changes to the orbit. This mapping (which assumes an impulse approximation) is derived using the fact that for orbits with sufficiently high eccentricity, the phase of a mode l at the time of periastron passage is essentially arbitrary.

2.4.1 Energy transfer

To simplify the following discussion, we assume that only star 1 is tidally active. In practice we calculate the energy transferred between each star and the orbit.

During chaotic evolution, the eccentricity may vary over a wide range of values and any one change in eccentricity after a periastron encounter can be up to an order of magnitude more than the change associated with energy transfer to a quiescent star. This reflects the fact that the energy transferred to the tides depends on the amount of tidal energy already present as well as the oscillatory state of the stars and can be understood as follows. Let δE_{osc} be the energy that would be transferred to the polytrope over an orbital period were no tidal energy present initially, and let E_{osc} be the tidal energy actually present initially. Also let δA_{osc} and A_{osc} be the amplitudes of oscillation associated with δE_{osc} and E_{osc} respectively. If $E_{\text{osc}}^{\text{new}}$ is the tidal energy after the subsequent periastron encounter, then (since the analysis is linear in the tidal amplitudes)

$$E_{\text{osc}}^{\text{new}} \sim (A_{\text{osc}} + \delta A_{\text{osc}})^2 \quad (31)$$

so that the energy transfer, ΔE_{osc} , is given by

$$\Delta E_{\text{osc}} \equiv E_{\text{osc}}^{\text{new}} - E_{\text{osc}} \sim 2A_{\text{osc}}\delta A_{\text{osc}} + \delta A_{\text{osc}}^2. \quad (32)$$

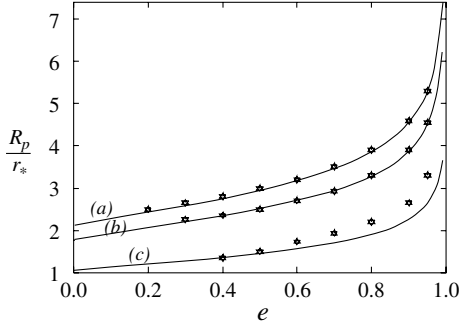


Figure 6. Chaos boundaries for (a) and (b) $n = 1.5$ polytropes for $q = 2$ and $q = 1$ respectively and (c) equal mass and $n = 3$. The stars are empirical data and the curves are from equation (28). The chaotic nature of a binary is determined by comparing two orbits with almost identical initial conditions.

In other words

$$\Delta E_{\text{osc}} = 2\sqrt{E_{\text{osc}}}\sqrt{\delta E_{\text{osc}}}\cos\gamma + \delta E_{\text{osc}}, \quad (33)$$

where γ is the relative phase of the orbit and the oscillation at periastron. Thus it is possible for energy to be returned to the orbit, as well as for the amount transferred to be significantly more than just δE_{osc} .

We will use this description to formulate a mapping which mimics the chaotic tidal interaction. Chaos is introduced by selecting a random value for γ in the interval $[0, 2\pi]$. To calculate δE_{osc} , we use a *modified* version of the formula given in Press & Teukolsky (1977) for the energy transfer from a parabolic orbit. Their formulation involves a similarity variable which depends on the mass ratio and the periastron separation; the present formulation generalizes this similarity variable to include eccentricity. While not exact, the prescription is of sufficient accuracy for the range of eccentricities relevant to chaotic orbits, including hyperbolic orbits. Appendix A gives a summary of the generalized Press and Teukolsky analysis. We take E_{osc} to be composed of contributions for the $l = 2$ and $l = 3$ f -modes which are the most energetic during close encounters. In the following, ω_l refers to the (unscaled) oscillation energy in the l th mode.

For chaotic orbits we include linear and non-linear contributions to the damping of the oscillations. As for normal tidal interactions, damping of modes is characterized by their Q value. We define a non-linear Q_{NL} for each star in terms of the linear value following Kumar & Goodman (1996):

$$Q_{\text{NL}} = \frac{Q_i}{\sqrt{E_{\text{osc}}/E_*}}, \quad (34)$$

where $E_* = Gm_1^2/r_1$ so that only when the tidal energy is a substantial fraction of the stellar binding energy does non-linear damping contribute to energy loss. Defining a mode damping time-scale for each star, τ_d , such that $\tau_d^{-1} = \tau_L^{-1} + \tau_{\text{NL}}^{-1}$, where $\tau_{\text{NL}} = 2\pi Q_{\text{NL}}/\omega_2$, the energy in this mode by the time of the next periastron encounter will have decayed by a factor $\exp(-T_{\text{orb}}/\tau_d)$, where T_{orb} is the orbital period. We assume that this time-scale is the same for the $l = 2$ and $l = 3$ f -modes, which are the only modes included in the mapping.

Using the superscript n to denote a quantity following the n th periastron encounter, the energy in mode l following the $(n + 1)$ th periastron passage is

$$E_l^{n+1} = r_l^n E_l^n + 2\sqrt{r_l^n E_l^n} \sqrt{\delta E_l} \cos\gamma + \delta E_l^{n+1}, \quad (35)$$

where

$$r_l^n = \exp(-T_{\text{orb}}^n/\tau_d). \quad (36)$$

2.4.2 Orbit

In order to calculate δE_l^{n+1} and hence E_l^{n+1} (see Appendix A), we require the periastron separation and the eccentricity which can be calculated as follows. Let E_0 be the total energy of the system initially, and let E_{orb}^n and $E_{\text{osc}}^n \equiv \sum_l E_l^n$ be the orbital binding energy and the tidal energy respectively during the n th orbit. Similarly let J_0 be the total angular momentum of the system, and let J_{orb}^n and $J_{\text{osc}}^n \equiv \sum_l J_l^n$ be the orbital angular momentum and tidal angular momentum respectively during the n th orbit, where $J_l^n = \alpha_l E_l^n$ and $\alpha_l = l/\omega_l$ (Lai, Rasio & Shapiro 1994). If $E_{\text{dec}} = \sum_n \sum_l (1 - r_l^n) E_l^n$ is the total energy lost from the system since the beginning of the chaotic phase, then (taking the interaction energy between the tides and the orbit to be negligible away from periastron)

$$E_{\text{orb}}^n + E_{\text{osc}}^n = E_0 - E_{\text{dec}}. \quad (37)$$

Since the total angular momentum of the system is conserved, we have

$$J_{\text{orb}}^n + J_{\text{osc}}^n = J_0. \quad (38)$$

In terms of the scaled semimajor axis, a_n , and the eccentricity, e_n , of the n th orbit, the orbital binding energy and orbital angular momentum are

$$E_{\text{orb}}^n = -\frac{1}{2} C_E q / a_n \quad (39)$$

and

$$J_{\text{orb}}^n = C_J \left[\frac{q^2}{1+q} a_n (1 - e_n^2) \right]^{1/2} \quad (40)$$

respectively, where $C_E = Gm_1^2/r_1$ and $C_J = \sqrt{Gm_1^3 r_1}$. Equations (37) and (39) give the semimajor axis:

$$a_n = -\frac{1}{2} C_E / (E_0 - E_{\text{dec}} - E_{\text{osc}}^n), \quad (41)$$

while equations (38) and (40) give the eccentricity:

$$e_n = \left[1 - \frac{(1+q)(J_0 - J_{\text{osc}}^n)^2}{q^2 C_J^2 a_n} \right]^{1/2}. \quad (42)$$

The periastron separation is then given by

$$R_p^n = a_n (1 - e_n). \quad (43)$$

Fig. 7 compares the results of integrating the full dynamical equations (Mardling 1995b) with those using the mapping described above for an initially parabolic orbit with $R_p = 3r_*$, where r_* is the radius of the extended star. The qualitative behaviour is similar. A system remains chaotic until it has lost sufficient energy to cross the chaos boundary.

2.4.3 The chaos boundary and damping

For a given binary, chaotic behaviour is only possible for a range of eccentricities. As the binary loses energy, this range decreases until it crosses the chaos boundary permanently. This process is illustrated in Fig. 3 for a binary which starts with an eccentricity $e_0 = 1$. Just as there exists a chaos boundary corresponding to zero initial tidal energy (as in Fig. 2), there also exist boundaries

for non-zero tidal energies which lie above the zero energy boundary (see Mardling 1995a). These boundaries govern the minimum eccentricity a chaotic binary can acquire. Thus we need to prescribe this minimum eccentricity which will increase with the amount of energy the binary has dissipated since the beginning of the chaotic phase.

Fig. 8 illustrates the path an evolving binary takes as it loses energy. In the absence of dissipation, the binary would simply move along the curve $\overline{ab}$. As the system loses energy, however, the maximum eccentricity will lie on the curve $\overline{ac}$ and the minimum on the curve $\overline{bc}$. We first need to specify the point $\mathbf{c}$ given initial values $R_p = R_p^0$ and $e = e_0$. This point is the intersection of the curve for which the orbital angular momentum is constant, i.e. $R_p(1+e) = R_p^0(1+e_0)$, and the chaos boundary given by equation (28). Writing these two equations as $p = R_p/r_1$ in terms of e and equating gives an implicit function of e as well as $p_0 = R_p^0/r_1$, e_0 , q and ω :

$$\frac{(1+e)^{2.4}}{(1-e)} = \left(\frac{p_0(1+e_0)}{C} \right)^5 \left(\frac{\omega^2}{1+q} \right)^2. \quad (44)$$

To avoid solving equation (44) numerically, we fit a curve to the inverse of the left-hand side as follows. For $0 \leq e \leq 0.99$ calculate

$$x = \ln \frac{(1+e)^{2.4}}{(1-e)}, \quad (45)$$

then find c_n such that

$$e = \sum_{n=0}^5 c_n x^n \quad (46)$$

(see Table B2 of Appendix B). The point $\mathbf{c} \equiv (p_c, e_c)$ is then given by

$$e_c = \sum_{n=0}^5 c_n \left\{ \ln \left[\left(\frac{p_0(1+e_0)}{C} \right)^5 \left(\frac{\omega^2}{1+q} \right)^2 \right] \right\}^n \quad (47)$$

and

$$p_c = p_0 \frac{1+e_0}{1+e_c}. \quad (48)$$

This procedure was checked by finding the root of equation (44) numerically for various values of the right-hand side and comparing to equation (47). The results are in excellent agreement.

The point $\mathbf{b}$ is related to the point $\mathbf{c}$ as follows. To get from $\mathbf{b}$ to $\mathbf{c}$, the system must lose an amount of energy equivalent to the difference in orbital energy between $\mathbf{a}$ and $\mathbf{c}$, that is $E_0 - E_{\text{orb}}(\mathbf{c})$. Given that at some arbitrary stage in the chaotic phase, the system has lost an amount of energy E_{dec} and the minimum eccentricity it can attain corresponds to the point $\mathbf{x}$, we have (see fig. 13 of Mardling 1995b)

$$E_{\text{orb}}(\mathbf{x}) = E_{\text{orb}}(\mathbf{b}) + E_{\text{dec}}. \quad (49)$$

Thus we have

$$E_{\text{orb}}(\mathbf{c}) = E_{\text{orb}}(\mathbf{b}) + [E_0 - E_{\text{orb}}(\mathbf{c})], \quad (50)$$

so that

$$E_{\text{orb}}(\mathbf{b}) = 2E_{\text{orb}}(\mathbf{c}) - E_0. \quad (51)$$

The point $\mathbf{b}$ satisfies both

$$E_0 = E_{\text{orb}}(\mathbf{b}) + E_{\text{osc}}(\mathbf{b}) \quad (52)$$

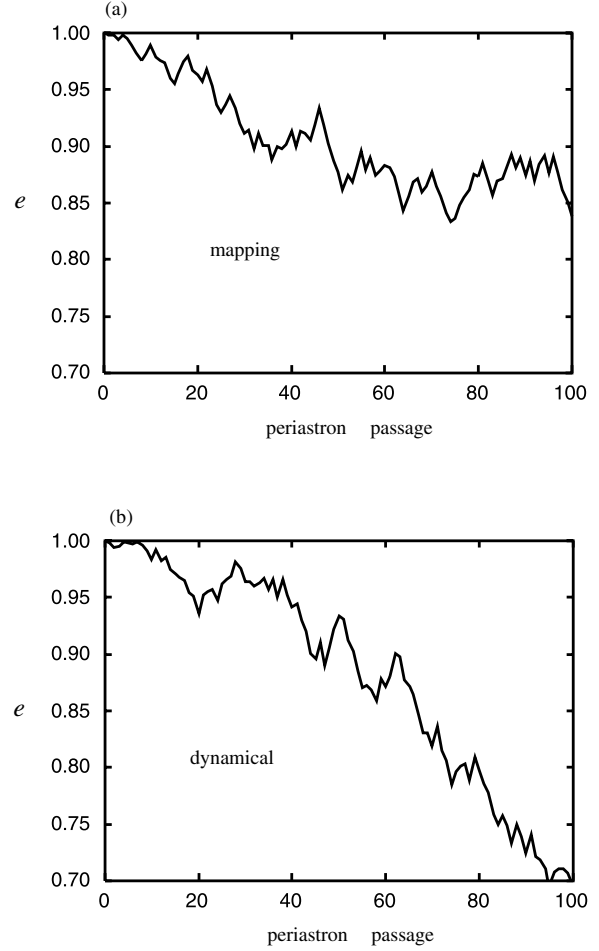


Figure 7. Comparison of chaotic behaviour for (a) mapping and (b) direct integration and (equation 42). Typical changes in the eccentricity and the erratic behaviour are modelled well.

and

$$J_0 = J_{\text{orb}}(\mathbf{b}) + J_{\text{osc}}(\mathbf{b}) = J_{\text{orb}}(\mathbf{b}) + \alpha_2 E_{\text{osc}}(\mathbf{b}), \quad (53)$$

where $\alpha_2 = 2/\omega_2$, with ω_2 being the $l = 2$ mode of the star which determines the chaos boundary. Eliminating $E_{\text{osc}}(\mathbf{b})$ from these equations gives

$$J_{\text{orb}}(\mathbf{b}) = J_0 - \alpha_2 (E_0 - E_{\text{orb}}(\mathbf{b})). \quad (54)$$

The semimajor axis a_b corresponding to point $\mathbf{b}$ comes from equations (39) and (51):⁴

$$\frac{1}{a_b} = \frac{2}{a_c} - \frac{1}{a_0}, \quad (55)$$

where $a_c = p_c/(1-e_c)$ and $a_0 = p_0/(1-e_0)$, while the eccentricity comes from equations (40), (51) and (54):

$$e_b^2 = 1 - \frac{1+q}{q^2} \frac{1}{a_b} \left\{ \frac{J_0 - \alpha[E_0 - E_{\text{orb}}(\mathbf{b})]}{C_J} \right\}^2. \quad (56)$$

The curve $\overline{bc}$ is given approximately by

$$p = Ae + B, \quad (57)$$

⁴If $E_0 > 0$, then we define $a_0 < 0$.

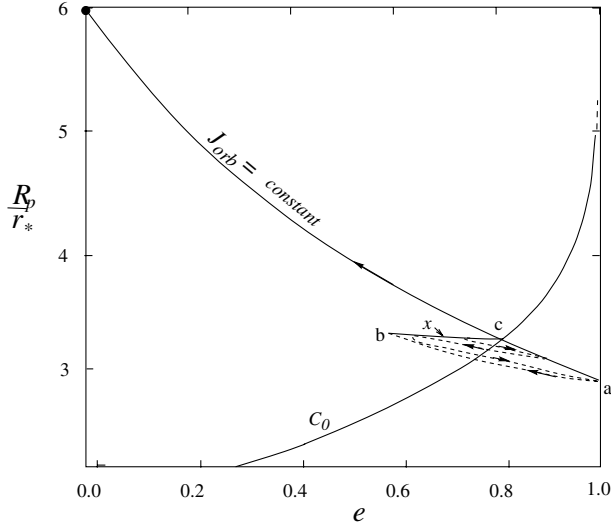


Figure 8. The evolutionary path of a damped chaotic binary. Initially the binary moves chaotically between points **a** and **b**. As it loses energy the length of this path shrinks and moves towards **c** at which point the binary ceases chaotic motion and evolves to the left along the curve $J_{\text{orb}} = \text{constant}$.

where

$$A = \frac{p_b - p_c}{e_b - e_c} \quad (58)$$

and

$$B = \frac{e_b p_c - e_c p_b}{e_b - e_c}. \quad (59)$$

This line may be parametrized by E_{dec} using equations (49) and (51) (and equation 39), i.e.

$$p = \frac{1}{\varepsilon}(e - 1), \quad (60)$$

where

$$\varepsilon = \frac{2}{a_c} + \frac{2}{C_E}(E_{\text{dec}} - E_0) \quad (61)$$

Equations (57) and (60) can then be equated and solved for the minimum eccentricity, e_{min} , as a function of E_{dec} , giving

$$e_{\text{min}} = \frac{1 + \varepsilon B}{1 - \varepsilon A}. \quad (62)$$

In practice, the mapping procedure described above (equations 35, 42 and 43) is implemented while $e_n > e_{\text{min}}$. If $e_n < e_{\text{min}}$, we set $\gamma = \pi$ temporarily in equation (35) which ensures that $e_{n+1} > e_{\text{min}}$. This mimics reflection off the chaos boundary. We proceed until $E_{\text{dec}} = E_0 - E_{\text{orb}}(\mathbf{c})$, after which the procedure for normal circularization is implemented.

Fig. 3 was produced using the procedures outlined in this section including the ‘normal’ phase. Thus the mapping is capable of calculating the non-chaotic part of the evolution as well if the procedure is continued after $E_{\text{dec}} = E_0 - E_{\text{orb}}(\mathbf{c})$.

2.4.4 Tidal disruption

Since chaotic orbital evolution can involve huge tidal energies, we need to specify a minimum periastron separation as a function of eccentricity, mass ratio and stellar type below which at least one

star can be considered to have disrupted tidally. Kochanek (1992) defines a disruption criterion according to whether or not a star is disrupted during the first few orbits following a parabolic encounter, his stars being modelled as irrotational Riemann ellipsoids. According to this criterion, a star will be disrupted if

$$\frac{R_p}{r_1} < 1.3 \left(\frac{m_1 + m_2}{2m_1} \right)^{1/3}, \quad (63)$$

where r_1 is the largest radius. Kochanek (1992) has a second more stringent disruption criterion which is based on the assumption that the binary can circularize fully before any significant damping has taken place. Such a scenario involves huge tidal energies once the binary is circular, and since the angular momentum in the tides is proportional to the energy in the tides and angular momentum is conserved, the minimum circular separation below which there are no stable solutions is considerably higher than for the case where the binary has time to damp the tides. In the latter case, the angular momentum transferred to the star(s) from the orbit will be just that which allows them to spin synchronously. The second criterion is

$$\frac{R_p}{r_1} < 1.7 \left(\frac{m_2}{m_1} \right) + 0.5 \left(\frac{m_1 + m_2}{2m_1} \right)^{1/2}. \quad (64)$$

Since the chaos boundary limits the tidal energy and angular momentum a star can absorb, we employ a disruption criterion which reflects this. This gives minimum separations which are greater than those given by equation (63) since there will be binaries which survive the first few orbits but will go on to be disrupted tidally as the tides are built up further through chaotic interaction with the orbit. On the other hand, since the tidal energy is restricted by the existence of a minimum eccentricity, this criterion will give minimum separations which are *less* than those given by equation (64). In the absence of a model which can simulate the behaviour of binaries which disrupt tidally after more than a few orbital periods, we define the new disruption criterion to be such that only binaries with $e_b > 0$ survive (see equation 56 and Fig. 8). In other words, binaries do not survive if the chaotic region extends all the way to circular orbits. This in some ways is analogous to the second Kochanek condition, but is less stringent because it does not assume that *all* orbits can do this. In addition, this criterion can be applied to any binary which is chaotic, not just capture binaries. The following empirical formula gives a value for e_c^{dis} , the value of e_c which corresponds to $e_b = 0$. One can show that it is independent of q and ω , that is, it does not depend on which chaos boundary is being considered. One calculates

$$e_c^{\text{dis}} = 0.079 - 0.022e + 1.275e^2 - 0.670e^3 \quad (65)$$

and compares it to e_c (equation 47). If $e_c > e_c^{\text{dis}}$, one proceeds with calculating the chaotic evolution. The implementation is described in Section 3.3.

2.5 Tidal capture

Although the star clusters we model are of moderate size, the possibility of binary formation via tidal capture exists. We have observed this in approximately one in 10 models, as well as direct collisions at about three times this frequency. As may be anticipated, successful tidal captures occur for pairs of stars with eccentricities very close to unity (a typical value, say, 1.0001),

corresponding to moderate relative velocities at infinity. In this process, the excess energy of the unbound hyperbolic motion of two stars is taken up by the tides of the stars when they pass through the pericentre of their relative orbit, resulting in a bound eccentric orbit. Given the structure of the stars via ω_i and Q_i , their radii and their mass ratio, q , it is possible to calculate the maximum periastron separation necessary for tidal capture to occur (Mardling 1996b). Fig. 9 shows the dependence of this as a function of q for polytropes of index 1.5 and for various velocities at infinity. These functions are conveniently represented by a single fitting function. If the periastron separation of a hyperbolic pair falls below $R_p^{\max}$, the pair is considered to be a capture and the ensuing chaotic evolution is calculated. The fitting formula is given by

$$R_p^{\max}/r_1 = (P_1(q, \tilde{v}_\infty) + (r_{21}^5/q^2)P_2(q, \tilde{v}_\infty))^{1/6}, \quad (66)$$

where

$$P(q, \tilde{v}_\infty) = (a_0 + a_1 q + a_2 q^2 + a_3 q^3)/\tilde{v}_\infty^\beta, \quad (67)$$

with $\tilde{v}_\infty = v_\infty/\text{km s}^{-1}$, v_∞ being the relative velocity of the stars at infinity, and different a 's and β 's for $n = 1.5$ and $n = 3$ polytropes (see Appendix B).

2.6 Stellar evolution

A binary can become tidally active once the radius of one of the stars grows during its post main-sequence evolution. Although the convective envelope of a giant can be modelled by a polytropic structure of index 1.5, a significant fraction of the mass of the star resides in the core so that its ability to absorb tidal energy is reduced. The star behaves tidally as a more centrally condensed star so that it becomes desirable to define an *effective* polytropic index.

To do this we make use of a calculation by McMillan, Taam & McDermott (1990a) for the normal modes of oscillation of a $0.8 M_\odot$ star at various stages of its ascent of the giant branch. As a first approximation, we assume that the dependence of the oscillation frequencies on the core mass is the same for all giants, independent of their mass and metallicity (the McMillan et al. study assumes low metallicity characteristic of Population II stars in globular clusters). Using the data from that study, we derive a cubic polynomial fit for the scaled oscillation frequencies of the $l = 2$ and $l = 3$ modes versus core mass over mass, as shown in Fig. 10(a). Points for which $m_c/m = 0$ correspond to the $l = 2$ f -modes of an $n = 1.5$ polytrope. The point corresponding to $m_c/m = 1$ comes from an extrapolation of Q versus ω to $Q = 0$ (Fig. 10b). Since the value $\omega = 7$ corresponding to $Q = 0^5$ is somewhat arbitrary, we did not obtain a good fit if we used the points corresponding to $m_c/m = 0.6$. These were left out of the fitting procedure but are shown nevertheless to indicate the increasing dependence of ω on m_c/m .

As the ratio of core mass to total mass increases, the ability of the envelope to store tidal energy (for a fixed value of p) decreases. Since the circularization time-scale is proportional to w^4/Q^2 (equations 11 and 25), it increases without bound if the core mass approaches unity. Thus there exists a competition between the circularization time-scale and the stellar evolution time-scale; it is possible in some cases for mass transfer to start before the binary has finished circularizing.

⁵ In fact we take this to be $Q = 0.0001$.

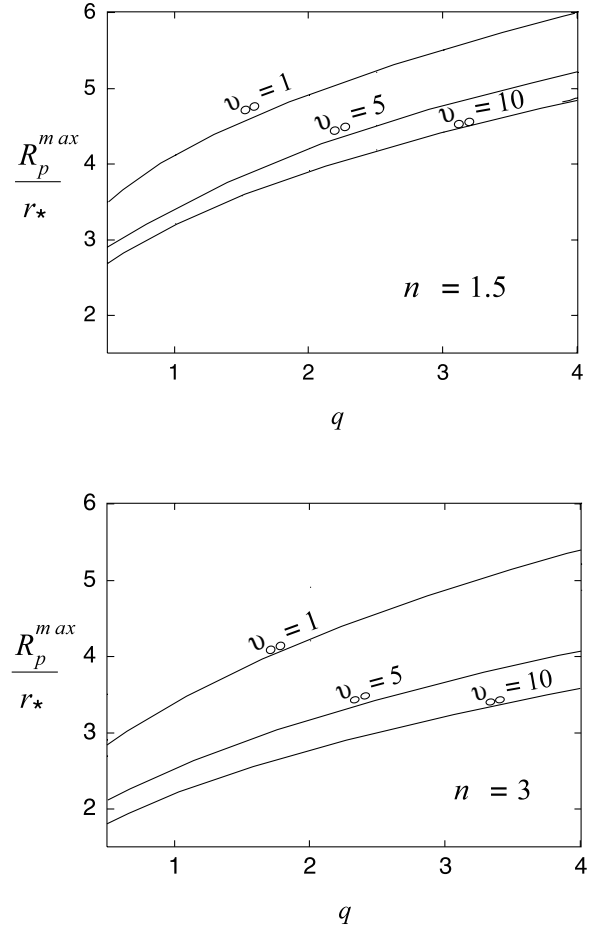


Figure 9. Maximum periastron separations for capture as functions of mass ratio and relative velocity (in km s^{-1}) at infinity.

This will also depend on the Q values of the stars, for which we follow the analysis of Verbunt & Phinney (1995). Comparing their equation (1) with our equations (11) and (18), and identifying the quantity m_{env}/m with the apsidal motion constant where $m_{\text{env}} = m - m_c$ is the mass in the giant's envelope, we can define the Q value of the star in terms of its luminosity $\mathcal{L}$:

$$Q \sim \frac{\sqrt{Gmm_{\text{env}}^{1/3}}}{r_*^{5/6}\mathcal{L}^{1/3}}. \quad (68)$$

The luminosity itself is obtained from the core mass-luminosity relation (Tout et al. 1997) by

$$\mathcal{L} = 1.98 \times 10^5 m_c^6. \quad (69)$$

The effective polytropic index n (which is not used, but is of interest none the less), and the corresponding (scaled) frequencies ω_l and overlap integrals Q_l for $l = 2, 3$ are calculated as follows:

$$\omega_l = a_{0,l} + a_{1,l}\sigma + a_{2,l}\sigma^2 + a_{3,l}\sigma^3, \quad (70)$$

$$Q_l = b_0 + b_1\omega_l + b_2\omega_l^2 + b_3\omega_l^3, \quad (71)$$

$$n = c_0 + c_1\omega_2 + c_2\omega_2^2, \quad (72)$$

where the coefficients are listed in Table B4 of Appendix B.

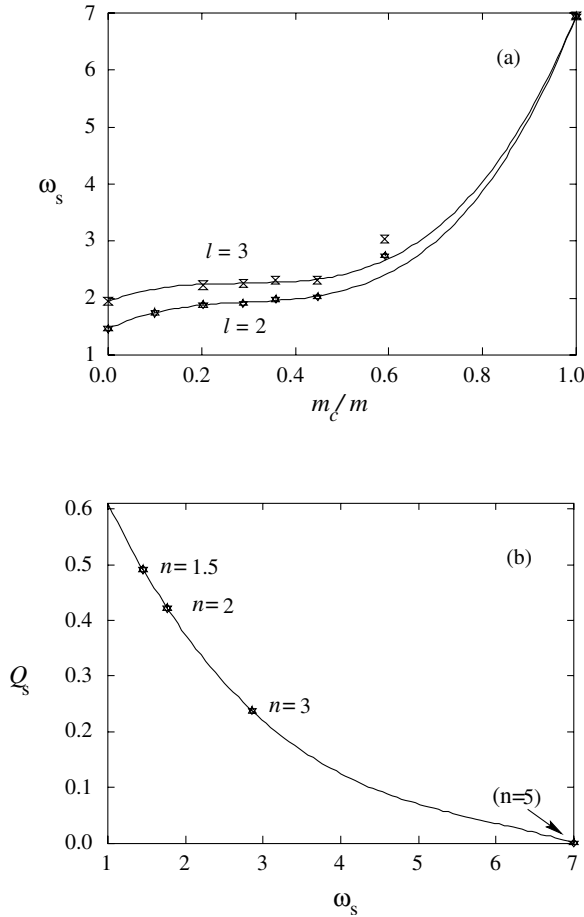


Figure 10. Interpolation for giant evolution. Here ω_s and Q_s are the scaled mode frequency and overlap integral.

3 N-BODY IMPLEMENTATION

A realistic simulation of star clusters requires the combination of an efficient N -body code with astrophysical procedures for studying a variety of processes. The most relevant features such as mass loss from stellar winds, supernova events, tidal dissipation and Roche-lobe overflow have already been summarized elsewhere (Aarseth 1996), together with the integration methods. In the present paper, we are mainly concerned with the implementation of tidal dissipation based on consistent stellar evolution.

In the following subsections, we describe the main algorithms for dealing with tidal circularization and chaotic motions of binaries in the code NBODY4 (Aarseth 1996). Although implemented on the HARP special-purpose computer, some of this treatment together with instant circularization has also been included in the codes NBODY5 and NBODY6 which are intended for standard workstations or vector supercomputers (Aarseth 1994). The use of the powerful HARP computer facilitates simulations of large open cluster models containing significant binary populations. In the subsequent discussions it will be assumed that the orbits of single stars and the centres of mass of binaries are advanced consistently, which leaves us free to concentrate on aspects connected with binaries.

3.1 Initial binary distribution

The addition of primordial binaries has a significant effect on the

dynamical evolution of star clusters (McMillan, Hut & Makino 1990b, 1991; Heggie & Aarseth 1992). This introduces further uncertainties concerning the choice of initial conditions but we may be guided by observational data and theoretical considerations. In particular, the semimajor axis distribution and initial mass function (IMF) are likely to influence the results strongly. In the present astrophysical simulations, additional effects relating to mass loss and Roche-lobe overflow also complicate the situation. Here we adopt the approach of using observationally based parameters. As a standard model, we choose an initial fraction $f_b = 0.05$ of relatively hard binaries for an open star cluster model containing 9500 single stars and 500 binaries, making a total membership of $N = 10\,500$ (Aarseth 1996). An initial King-type model for tidally truncated clusters with concentration parameter $W_0 = 5$ is generated by the algorithm of Heggie & Ramamani (1995), with half-mass radius $R_h \approx 3$ pc.

The IMF is obtained from Kroupa, Tout & Gilmore (1993) with the mass generating function $M(X)$ given by

$$M(X) = 0.08 + \frac{\gamma_1 X^{\gamma_2} + \gamma_3 X^{\gamma_4}}{(1 - X)^{0.58}}, \quad (73)$$

where X is a uniformly sampled random number, $\gamma_1 = 0.19$, $\gamma_2 = 1.55$, $\gamma_3 = 0.05$ and $\gamma_4 = 0.6$. We adopt masses in the range $15.0\text{--}0.2 M_\odot$ as a realistic representation of young open clusters. Binary component masses m_i, m_j are first chosen independently from this distribution. In order to simulate an observational bias towards more equal mass ratios, the actual mass ratio $q = m_i/m_j$ is modified by the correlation $q' = q^{0.4}$, subject to the sum remaining constant (Eggleton, private communication). This procedure yields component masses of 7.3 and $1.8 M_\odot$ for the heaviest binary in one model, compared to $14.4 M_\odot$ for the most massive single star. Note that with a few exceptions, the heaviest binary masses tend to be somewhat lighter than the corresponding single stars here because of the smaller statistical sample. This property of the adopted IMF has interesting consequences for mass segregation since the roles are reversed once significant stellar evolution has taken place.

The initial period distribution is based on a complete collection of all observational data of main sequence and pre-main sequence binaries (Kroupa 1995), where we have imposed a lower limit of $P_{\min} \approx 4$ d. Currently there is also an upper limit of $a_{\max} \approx 50$ AU on the semimajor axis for technical reasons, compared to a typical hard binary of mean mass components with $a_{\text{hard}} \approx 2R_h/N \approx 120$ au.

Initial eccentricities are selected at random from the thermal distribution $f(e) = 2e$. However, we adopt provisional enhanced pre-main sequence radii given by $r_i/r_\odot = 5(M/M_\odot)^{1/2}$ in order to model the early stage of contraction and exclude possible collisions. Consequently, binaries for which $a(1 - e) < \max(r_i, r_j)$ are rejected because collision would have occurred. The additional process of pre-main sequence ‘eigenevolution’ (Kroupa 1995) has also been implemented in the code, although the model described in this paper did not include this effect. Typically, about 25 collision candidates are rejected, leaving some 10 binaries with periods below 20 d. Finally, the starting epoch is taken to be zero and all radii are therefore initialized on the zero-age main sequence using the fast look-up procedure for Population I stellar metallicity (Tout et al. 1997).

3.2 Dynamical treatment

A typical star cluster simulation extends over several billion years,

whereas the shortest binary periods may be just a few days or even hours. We therefore need an efficient scheme for advancing the orbital motions of binaries over the cluster lifetime while retaining a consistent description. The dynamical evolution of hard binaries is characterized by long durations of extremely small perturbations interspersed by shorter intervals of high dynamical activity which may change the binding energy significantly or even lead to disruption. In addition to the time-scale problem, the small pericentre separation of eccentric binaries may also present severe numerical difficulties for individual orbit computations based on brute force methods, unless relative coordinates are introduced in some other way.

We exploit the dominant two-body motion by adopting a formulation based on the relative separation; this is especially convenient for the tidal circularization formulation which requires the identity of the binary components to be known. The classical Kustaanheimo–Stiefel (1965, hereafter KS) regularization transforms the singular two-body variables into a set of well-behaved equations of harmonic oscillator type which are independent of the eccentricity. The corresponding equations of perturbed motion are given by

$$\mathbf{u}'' = \frac{1}{2}h\mathbf{u} + \frac{1}{2}R\mathcal{L}^T\mathbf{P}, \quad (74)$$

$$h' = 2\mathbf{u}' \cdot \mathcal{L}^T\mathbf{P}, \quad (75)$$

$$t' = \mathbf{u} \cdot \mathbf{u}. \quad (76)$$

Here $\mathbf{u}$ denotes the new coordinates which satisfy the relation $\mathbf{u} \cdot \mathbf{u} = R$ for the separation R , h is the two-body binding energy per unit reduced mass, $\mathcal{L}^T(\mathbf{u})$ is the transpose of a linear 4×3 matrix and $\mathbf{P}$ is the perturbation (in physical units) exerted by the other particles. In this treatment, the time transformation (76) provides a relation between the fictitious time τ and physical time t , whereas the complete orbital solution is obtained by integrating the corresponding centre-of-mass motion represented by an additional pseudo-particle.

The semimajor axis and eccentricity of the two-body motion play a major role in the present formulation. These quantities can readily be determined from the KS variables by the simple relations

$$a = -\frac{m_b}{2h}, \quad (77)$$

$$e^2 = \left(1 - \frac{R}{a}\right)^2 + \frac{t'^2}{m_b a}, \quad (78)$$

where m_b is the mass of the binary and $t'' = 2\mathbf{u}'' \cdot \mathbf{u}$. In the following sections, e denotes the integrated orbital eccentricity (78), whereas subscripted quantities (e_p and e_t) refer to eccentricities modified by tidal circularization.

Decision-making for existing KS solutions is mainly based on the relative perturbation, $\gamma = PR^2/m_b$ which is formed after each integration step. During the integration, two new particles are selected for KS treatment if their time-steps are suitably small and the two-body motion is dominant, irrespective of the sign of h . The KS scheme has been generalized to an arbitrary number of two-body systems and is therefore suitable for studying star clusters containing a substantial population of primordial binaries. Further discussions of the KS implementation and the relevant transformations can be found elsewhere (Aarseth 1994). For some time now, the equations of motion (74) and (75) have been integrated by the Hermite scheme in order to preserve formal similarity with the treatment of single particles on the HARP-2

special-purpose computer (Aarseth 1996). A very recent development is the implementation of Stumpff functions in the Hermite integration scheme (Mikkola & Aarseth 1998) which ensures higher accuracy for small perturbations. Moreover, the so-called ‘slow’ KS procedure (Mikkola & Aarseth 1996) utilizes the principle of adiabatic invariance for small perturbations, such that one KS orbit may in fact represent a number of physical orbits.

In the case of small external perturbations (say $\gamma < 10^{-6}$ at the apocentre), the osculating two-body elements are assumed to be constant over one or more periods, leaving only the centre of mass to be advanced as for single particles. The actual time interval for assuming a dormant binary can be estimated by considering the relative motion of the nearest neighbours, but is in any case limited to the time-step of the associated centre-of-mass particle. Once a binary has been identified by its dominant interaction, it is treated as a dynamical entity until one of the following conditions arise: (i) an intruding particle approaches closely and replaces one of the components, (ii) it becomes unbound owing to mass loss, or (iii) the binary forms a stable triple or quadruple hierarchical system, according to some stability condition (discussed in Section 4.1). In the third case, the binary is replaced by its centre of mass but the elements are preserved and used to re-initialize the orbit if hierarchical stability is no longer satisfied (strong external perturbation or small pericentre distance). This procedure enables the outer orbit to be treated as a KS solution in the usual manner, thereby speeding up the calculation further. Such hierarchical systems may be regarded as newly formed binaries, with the inner pair acting as a massive point-mass primary for dynamical purposes. Examinations of star cluster simulations show that several stable hierarchies may exist simultaneously (Kiseleva et al. 1996; de la Fuente Marcos et al. 1997), and this behaviour is confirmed in the present models.

The case of a single particle or binary interacting strongly with another binary is of considerable interest dynamically. Provided they are sufficiently compact, such systems are studied by the chain regularization method (Mikkola & Aarseth 1993) which is based on a generalization of KS to arbitrary memberships and also includes external perturbations (Aarseth 1996). For practical reasons, the treatment is limited to subsystems with 3–6 members. The chain method has recently been modified to incorporate the principle of slow-down (Mikkola & Aarseth 1996). Depending on the strength of the perturbation, the effective period of any energetic binary inside a subsystem is increased; hence the shared time-step is not tied to the original Kepler period. This formulation permits a consistent treatment of binaries with arbitrarily short periods interacting with other members in small clumps, instead of assuming unperturbed two-body motion during this stage. Since the chain regularization is relatively expensive because of the high-order integrator, this powerful technique is only used for a few hundred critical cases in a typical cluster model (compared to some $\sim 2 \times 10^4$ new KS treatments for elliptic or hyperbolic orbits). However, many of these events involve very large energy changes which may otherwise cause severe numerical problems. The ability to perform *in situ* corrections for astrophysical effects in such subsystems also presents significant advantages over direct integration methods.

3.3 Decision-making

The present scheme distinguishes between single stars and binaries, where the latter require special attention. In the former case it is sufficient to update the stellar mass and radius of each

member at frequent time intervals as the integration proceeds. The binary motions are described by a two-body formulation which provides direct information about the orbital elements. Having ready access to the natural variables therefore enables several important processes to be implemented without additional programming complications.

The synthetic stellar evolution scheme employs a fast look-up of the radius, luminosity (L), core mass (m_c) and evolution type as a function of time (Eggleton, Fitchett & Tout 1989; Tout et al. 1997), using suitably small intervals for the different characteristic stages to model the changes. We assign two additional parameters to each star in order to facilitate the decision-making; the next epoch for updating the stellar parameters (radius or mass), T_{ev} , and an evolution index, K , which characterizes the current stellar type. Here $K = 0, 1, \dots, 12$ represents the different stages from low-mass main-sequence stars to neutron stars, black holes and the products of collisions. The times T_{ev} are updated at frequent intervals for each stage of the Hertzsprung-Russell (HR) diagram, such that the radius changes in a smooth manner (Aarseth 1996). Evolving stars are assumed to lose mass instantaneously by stellar winds according to the modified Reimers expression (Tout, private communication)

$$\frac{dm}{dt} = \frac{4 \times 10^{-13} rL}{m} \quad (79)$$

in units of $M_\odot \text{yr}^{-1}$, where a suitable time average of rL is used. Massive stars (above $\approx 7 M_\odot$) terminating their evolution are defined as neutron stars or black holes, depending on whether the core mass lies in the range $[1.4, 1.6] M_\odot$ or exceeds $1.6 M_\odot$, otherwise a white dwarf is formed.

It is also convenient to identify the evolutionary state of a binary by defining a corresponding index for the centre of mass, with $K = 0$ representing an initial binary which may either become chaotic ($K = -1$) or begin tidal circularization ($K = -2$) ending in a circular orbit ($K = 10$) and may later undergo two or more stages of Roche-lobe overflow ($K = 11-14$). In this paper, we are mainly concerned with processes involving tidal dissipation, whereas the Roche treatment is discussed elsewhere (Tout et al. 1997).

There are several ways in which an existing binary may begin a tidal interaction. First, the orbital eccentricity may increase by external perturbations. Since the semimajor axis tends to remain constant for weak perturbations, this results in a decrease of periastron separation. Second, the radius tends to grow with time for non-degenerate stars (except for the helium-burning stage) and may therefore satisfy the condition for tidal interaction, unless the compensating effect of mass loss with consequent orbital expansion intervenes. It is desirable that the increase of stellar radius is relatively small (say ≤ 3 per cent) at each updating because of the need for a gradual transition to tidal dissipation. Another interesting situation may arise after a strong three-body interaction in which the remaining binary acquires a high eccentricity ($e \approx 0.99$), frequently resulting in chaotic motion. High eccentricities may also arise more gradually if the binary forms part of a hierarchical triple and the inclination angle of the orbital planes is high (Heggie, private communication).

The decision-making is based on equation (65), where we distinguish the three relevant solution paths of chaotic and normal tidal dissipation, as well as a state which results in immediate collision.

Normal tidal dissipation is initiated in the following way. At each pericentre passage of perturbed binary motion, the

circularization time, t_{circ} , is evaluated, provided the periastron separation $R_p < 10 \max(r_i, r_j)$ and no collision is indicated. A binary is then selected for special treatment if $t_{\text{circ}} < 2 \times 10^9 \text{ yr}$. The question of whether Hertzsprung-gap stars (type 2, which are assigned a constant core mass) should be treated as giants is under consideration.

In the following discussion we first outline procedures for non-chaotic motion, which occurs if the orbit lies outside the chaos boundary. In addition to the evolution index $K = -2$ used for decision-making, the tidal circularization treatment employs three orbital parameters; the pericentre time t_0 , pericentre distance R_p and eccentricity e_p , first specified at the time when tidal dissipation is initiated. For unperturbed motion, these parameters define the entire subsequent orbital evolution (see Section 2).

In the case of unperturbed motion, the circularization time is checked as above at the next time for identifying approaching particles which may become significant perturbers (i.e. $\gamma_k > 10^{-6}$ owing to the mass m_k). Before initializing the pericentre parameters, the current KS variables are transformed from the usual apocentre position (which is convenient for checking or initializing perturbed motion) to the pericentre.

The less frequent case of a perturbed tidally circularizing binary is considered at each pericentre until the orbit has been circularized, whereupon the centre-of-mass evolution index is redefined to $K = 10$ as an indicator for the Roche stage to be checked during the updating of radii or masses. For convenience, a final eccentricity $e_{\text{circ}} = 0.002$ is adopted (rather than zero) in order to facilitate the definition of any subsequent perturbed apocentre or pericentre in terms of changes in the sign of the radial velocity, t'' , during one integration step.

We now turn to the case of chaotic motion which requires special treatment. This occurs if the orbit lies inside the chaos boundary. After the quantities E_b , J_b and the polytropic index n have been initialized, the formulation of Section 2.4 is implemented, resulting in new values of the binding energy and eccentricity. These procedures are employed at each subsequent pericentre passage. Hence the definition of unperturbed motion is suppressed during this stage which is usually short lived. Unless collision occurs, the chaos treatment is terminated when the system has lost enough energy to cross the chaos boundary permanently (see Section 2.4), followed by transition to normal tidal circularization.

Additional procedures are required to include the small but discontinuous changes which arise as a result of the increase of radius or mass loss. In the former case, a simple rectification of the pericentre parameters, R_p , e_p and t_0 , is carried out once the new elements have been obtained. This updating is also performed if an active KS binary becomes dormant (i.e. unperturbed). Following significant mass loss or increase of radius for the chaotic case, the chaos boundary must be re-evaluated and the nature of the motion assessed. As an example of complications which must be considered we mention a sudden mass loss as a result of neutron star formation, followed by a velocity kick (Aarseth & Mardling 1997). Such an event may produce a transition from non-chaotic to chaotic motion which could possibly explain some observed systems.

According to SPH simulations, two stars approaching each other within a certain separation may produce one object on a short time-scale. We employ the collision criterion (Kochanek 1992)

$$R_{\text{coll}} = 1.7 \left(\frac{m_b}{2m_k} \right)^{1/3} r_k, \quad (80)$$

where k denotes the largest stellar radius. The condition $R < R_{\text{coll}}$ does not automatically result in collision. Thus the actual outcome depends on the stellar type; in particular the core mass is used to evaluate the common-envelope stage (Tout, private communication; Tout et al. 1997) which may give rise to a short-period binary consisting of at least one degenerate star instead of a collision product. Note that for main-sequence binaries undergoing tidal circularization, such collisions do not usually occur until well after this stage has been completed because of the increasing pericentre distance. In the absence of a reliable collision criterion for evolved stars, we also use relation (80) for giants as a temporary measure.

3.4 Orbital corrections

The procedures outlined above require various corrections to be carried out. We first describe the treatment for KS variables which are ideal for this purpose.

In the case of normal tidal evolution, the predicted eccentricity e_f following the time interval $t - t_0$ is given by equation (23). For unperturbed two-body motion the corresponding pericentre distance is then

$$R_p = \frac{R_p^0(1 + e_p)}{1 + e_f}, \quad (81)$$

which assumes angular momentum conservation during the same interval. However, this expression is not appropriate for perturbed motion, and it is necessary to re-define the pericentre distance by replacing R_p^0 in (81) by the actual distance $R = \mathbf{u} \cdot \mathbf{u}$. Hence the eccentricity factors in (81) contain the modification resulting from circularization. Note that these factors play a minor role in rare cases of large perturbations since typical circularization times are long compared to the orbital period.

The relevant quantities for tidal interaction (R_p^0, e_p, t_0) are only updated in the case of perturbed motion or stellar evolution procedures (changes in mass or radius), since otherwise the current values can always be expressed in terms of the initial values. However, the corresponding KS elements are modified frequently to yield a consistent cross-section for dynamical interactions and perturber selection. Also note that the stellar radii may expand significantly during circularization involving late stages of evolution (i.e. red supergiants), hence the gradual increase of the pericentre by equation (81) may not be sufficient to prevent the collision condition (80) from being reached prematurely. In order to suppress this behaviour, which may be the result of inappropriate Q values, we check the Roche condition by temporarily setting $a = R_p(1 + e_f)$, which would result from angular momentum conservation. Consequently, if the primary is about to exceed its Roche radius, given by equation (5), we enforce circularization at the *reduced* semimajor axis. This procedure has also been adopted because a Roche treatment for eccentric orbits is lacking at present. However, the final outcome usually involves common-envelope evolution and is similar to the standard episodes of Roche-lobe overflow which terminate in coalescence.

After ensuring the exact pericentre phase by combining Kepler's equation with the analytical solution, corrections to the basic KS variables are performed, yielding new values by

$$\begin{aligned} \mathbf{u}_{\text{new}} &= c_1 \mathbf{u}_{\text{old}}, \\ \mathbf{u}'_{\text{new}} &= c_2 \mathbf{u}'_{\text{old}}, \end{aligned} \quad (82)$$

where $c_1 = (R_p/R)^{1/2}$ and $c_2 = 1/c_1$ by angular momentum conservation. In the more common case of unperturbed motion,

the corresponding change in the binding energy per unit reduced mass is obtained from angular momentum conservation which gives

$$\Delta h = \frac{m_b(e_f^2 - e^2)}{2a(1 - e^2)}, \quad (83)$$

where a and e are the current orbital parameters. In the alternative case of perturbed motion, the energy change since the previous updating can be written as

$$\Delta h = \frac{m_b e(e_f - e_p)}{a(1 - e^2)}, \quad (84)$$

where e_p is the orbital eccentricity at the previous pericentre passage. Hence we superimpose the predicted change in eccentricity over the last period as a result of tidal circularization, $\Delta e = e_f - e_p$, as a differential correction. In exceptional cases of large time intervals (i.e. hierarchical inactivity), the new energy is calculated explicitly from the corrected elements since otherwise the new pericentre may be inconsistent. An appropriate correction $\mu \Delta h$, with μ being the reduced mass, is accumulated in order to facilitate total energy conservation. Finally, the KS motion is re-initialized in the standard way if the orbit is perturbed.

KS correction procedures for chaotic motion are essentially similar to those above. Given the new binding energy $h = E_{\text{orb}}/\mu$ and eccentricity (equations 39 and 42), the second scaling factor takes the form

$$c_2 = \left(\frac{m_b + hR_p}{m_b + h_0R} \right)^{1/2}, \quad (85)$$

where we have made use of the explicit KS energy relation $h = (2\mathbf{u}' \cdot \mathbf{u}' - m_b)/R$, with h_0 denoting the old binding energy. The KS variables are then corrected as above, followed by initialization of the appropriate polynomials if the motion is perturbed.

Both types of tidal dissipation may be affected by mass loss owing to stellar winds and supernova events. In order to reduce the numerical effort owing to corrections, we delay the implementation of mass loss until the predicted amount exceeds 1 per cent. Any such mass loss is assumed to be instantaneous from the cluster and the total energy is adjusted accordingly in order to maintain conservation. For a KS binary undergoing gradual mass loss, we employ the adiabatic relation of $m_b a$ being constant. This gives rise to an increase in the specific binding energy by

$$\Delta h = \frac{\Delta m}{a} \left(1 - \frac{\Delta m}{m_b} \right), \quad (86)$$

where $\Delta m (>0)$ denotes the mass loss. Instead of using an instantaneous correction $\Delta h = \Delta m/R$, this procedure avoids an important bias (i.e. the pericentre point in the case of unperturbed tidal dissipation) by assuming that the mass is lost continuously along the orbit. After updating the binding energy, a homologous expansion is applied. This is achieved by adopting $c_1 = (a/a_0)^{1/2}$ to represent the coordinate transformation of (82), whereas the velocity scaling factor is given by $c_2 = (v^2/v_0^2)^{1/2}$, with the new regularized velocity in terms of *current* quantities

$$v^2 = \frac{1}{2}(m_b + hR). \quad (87)$$

Since $v^2 = -m_b(1 + e)/4$ for any KS solution at pericentre, and likewise $e = 1 - R/a$, it follows that $c_1 c_2 = 1$ by virtue of $m_b a$ being constant. Hence angular momentum is conserved and the eccentricity is also preserved, as has been verified numerically.

A corresponding orbital energy correction

$$\Delta E_b = -\mu \Delta h - h \Delta \mu \quad (88)$$

is accumulated, where the last term represents the change in reduced mass. In the case of stellar collisions, we perform corrections for any mass loss resulting from common-envelope evolution as well as the internal binding energy of the final binary components. These contributions are again accumulated in order to achieve energy conservation. Note that following mass loss, the reference values R_p^0 , e_p are updated by so-called rectification both before and after the tidal changes are applied, whereas the time t_p is still needed to evaluate the new eccentricity.

Following further code developments, the updating of stellar parameters (m , r) for the *inner* binary components of a hierarchical triple or quadruple can now be done *in situ* instead of a less elegant temporary termination and re-initialization. This is facilitated by modifying the (inactive) KS elements h , u , u' as well as the component masses (which are used for re-initialization). Note that mass loss from an inner binary tends to decrease the stability, which therefore needs to be re-checked. Additionally, the growth of the radius may signal the onset of tidal dissipation and hence require termination of the special KS treatment. Consequently, most hierarchies defined to be stable can now be treated as more permanent subsystems, rather than be subject to frequent terminations and re-initializations which tend to introduce spurious errors.

Several useful procedures connected with normal tidal evolution are included. Thus the evolution index K associated with the centre of mass is reset to zero (from -2) if $t_{\text{circ}} > 3 \times 10^9$ yr following the formation of a degenerate star which gives rise to considerable orbital expansion as a result of mass loss. Such a binary is said to be quiescent; it may begin another stage of circularization later after the second radius has expanded significantly or the eccentricity has increased. An orbit is defined to be circularized if the eccentricity falls below the nominal value for circular motion, $e_{\text{circ}} = 0.002$, and $K = -2$, whereupon its index is set to the circularized value $K = 10$. Conversely, a binary orbit satisfying $e > 0.1$ with $K = 10$ is treated as a standard case by setting $K = 0$. The latter is of considerable interest because this indicates that some nearby companion (usually in a hierarchy) has imparted significant perturbation and thereby contributed to overall relaxation by virtue of the binary's response.

Strong interactions studied by chain regularization may also lead to stages of normal tidal evolution or chaotic evolution. Such processes are treated similarly as discussed above, using the well-defined chain variables in order to avoid near-singular expressions. Several new technical problems must be faced in the chain description, which does not contain explicit expressions for the relevant two-body elements. Here the main task is to identify a given interaction as a likely candidate for special treatment. After identifying the closest particles in the chain with separation below a specified value, the periastron distance is determined by iteration.

In order to carry out the required modifications, we evaluate the semimajor axis and eccentricity by the following steps. First an iterative procedure is employed to obtain the chain variables at the pericentre passage epoch of the desired binary, as defined by a suitably small radial velocity. During this procedure, the pericentre distance R_p itself is evaluated by converting the chain variables to the KS form, combined with the appropriate time derivative. The relevant binding energy is obtained by subtracting all non-singular momentum and potential energy terms ($T + V$) from the total energy (E) of the subsystem, giving

$$E_b = E - T - V, \quad (89)$$

and hence the osculating semimajor axis is recovered from

$a = -0.5m_j m_i / E_b$. This enables the pericentre time to be calculated by Kepler's equation using the corresponding KS variables, and the iteration proceeds until convergence is achieved. Assuming the true pericentre position then finally yields the eccentricity from $e = 1 - R_p/a$. Although the high-order integrator only requires a small number of regularized steps for each binary orbit, the additional cost of such iterations (typically four or five steps) is negligible compared to the total effort.

Given the new two-body elements, the chain coordinates and momenta are modified accordingly, whereupon the integration proceeds with a consistent value of the total energy, together with analogous corrections as in the KS case. Thus each type of tidal dissipation may be initiated during the chain integration, with circularization occurring after termination if required since a new KS solution is assigned the appropriate value of the index K associated with the centre of mass. For example, a tidally active binary may undergo a strong interaction when studied by chain regularization, followed by exchange to another tidally active state and finally terminate as an appropriate KS solution. Likewise, physical collisions and possible mass loss through common-envelope evolution are also included in the chain regularization scheme. In this way, we achieve a uniform treatment of all relevant processes for hard binaries, irrespective of the solution method.

4 HIERARCHICAL SYSTEMS

N -body simulations of star clusters with primordial binaries frequently reveal hierarchical configurations which may remain stable for significant times. For example, a triple hierarchy may be formed when two binaries interact strongly, a process analogous to three-body exchange in which a resonant interaction between a single star and a binary results in the single replacing one of the components of the binary, with the displaced star escaping to infinity. In the binary–binary case, the most tightly bound binary (generally) plays the role of the single star in the triple case. Another way for a triple hierarchy to form is the classic ‘triple–binary’ mechanism in which two stars become bound when a third star carries away their excess energy of unbound motion. In the triple hierarchy case, a binary and a single become bound following the fly-by of a fourth star. This mechanism is much less efficient at forming hierarchical triples as it involves the coincidence of three objects (one of which is a binary), while the binary–binary mechanism involves only two.

Higher-order configurations (such as $2+2$ and $3+1$ quadruples) may also be formed in ways similar to those described above. Because their formation generally involves the triple exchange process, it should not be regarded as requiring the coincidence of n bodies, and hence to be highly improbable. Rather, because it involves the coincidence of two objects (be they individually binaries, triples or higher-order configurations), and because the more bodies involved in a stable configuration, the larger the cross-section it presents to passing objects, the probability of forming a high-order configuration is not negligible. In addition to these two factors, mass segregation increases the likelihood of encounters between these heavier objects, with mass loss from heavy single stars in the core increasing the relative mass of the hierarchies and hence their presence in the core.

In the case of an isolated cluster binary, the circularization process acts to reduce the eccentricity smoothly to zero and even small perturbations of a random nature do not in general change this tendency. Such binaries are by definition relatively close, with

a characteristic terminal semimajor axis of a few astronomical units for giant stars and often much less. This may be compared to the typical size of a hard binary, for which the semimajor axis $a_{\text{hard}} \approx 120 \text{ au}$ in our open cluster models. Hence there is ample *hierarchical space* for stable triples to exist, in which the inner component barely feels a significant perturbation.

Given that hierarchical triples with predominantly high outer eccentricity occur naturally in cluster simulations, it is desirable to model the subsequent evolution in other ways since the time-scale for direct integration is prohibitive.

4.1 Stability

For the following discussion, it is convenient to employ *hierarchical coordinates* to describe triple configurations. Thus for a system composed of an inner binary consisting of masses m_1 and m_2 , and a more distant third body of mass m_3 , we refer to the relative motion of the inner binary via e_{in} and a_{in} , the inner eccentricity and semimajor axis, while the motion of the third body relative to the centre of mass of the inner pair is referred to as the *outer* binary, with e_{out} and $R_{\text{p}}^{\text{out}}$ being the outer eccentricity and periastron separation. The inner and outer mass ratios are $q_{\text{in}} = m_2/m_1$ and $q_{\text{out}} = m_3/(m_1 + m_2)$ respectively.

Although the question of the ultimate stability of a ‘stable’ hierarchical triple is as yet unresolved, for most configurations (those which do not have extremely high outer eccentricities) there is a definite distinction between long and short term stability (meaning the escape of one of the components). The energy and angular momentum exchange processes are distinctly different for stable and unstable configurations. As with the binary-tides problem, there are two distinct types of tidal instability in the three-body problem: *tidal disruption* in which matter (stellar matter in the case of a binary and a single object in the triple case) is transferred or exchanged, and *chaotic energy exchange* which does not necessarily involve the transfer of matter. Fig. 11 illustrates the latter process. The inner semimajor axis of a stable triple is secularly constant⁶ (Fig. 11a), in contrast to an unstable system destined to lose one of its components (Fig. 11b). This is also reflected in the outer eccentricity. In the figure we compare the evolution of two initially similar configurations for which the inner eccentricity differs by one part in 10^5 . The unstable system (Fig. 11b) exhibits chaotic evolution as the orbits diverge after about five outer orbits. This behaviour is completely analogous to that in the binary-tides problem.

In analogy with the Roche problem, Zare (1977) derived a formula that describes which systems are capable of suffering exchange of a component between the inner and outer binary in a triple. While all such systems will inevitably involve the escape of one component, it is also possible for systems which energetically *cannot* involve exchange to lose a component. Until recently, no general stability formula for escape existed (apart from those derived empirically for restricted configurations: Harrington 1977; Eggleton & Kiseleva 1995). In Section 2.3, we derived an equation for the chaos boundary in the binary-tides problem (equation 28). In Mardling & Aarseth (1999), we show that stability against escape in the three-body problem is analogous to stability against chaotic energy exchange in the binary-tides problem, and derive a formula which is almost identical to equation (28), and which holds for arbitrary masses and inner and

outer eccentricities, although only for coplanar prograde motion. This expression is given by

$$\frac{R_{\text{p}}^{\text{out}}}{a_{\text{in}}} = C \left[(1 + q_{\text{out}}) \frac{1 + e_{\text{out}}}{(1 - e_{\text{out}})^{1/2}} \right]^{2/5} \equiv \frac{R_{\text{p}}^{\text{crit}}}{a_{\text{in}}}, \quad (90)$$

where $C = 2.8$ is determined empirically. Note that the inner eccentricity and mass ratio do not appear explicitly in this expression. Thus given e_{out} and q_{out} , a triple configuration with $R_{\text{p}}^{\text{out}}/a_{\text{in}}$ less than the value given by equation (90) can be expected to suffer the escape of one body. This formula holds for $q_{\text{out}} \leq 5$, beyond which all unstable systems can suffer *exchange* and hence be described by the exchange criterion of Zare (1977).

The decision-making for stable hierarchical systems makes full use of the KS description. A search for the dominant particle among the perturbers is made at each apocentre passage or alternatively at frequent intervals during unperturbed motion, if the corresponding centre-of-mass time-step is suitably small. Note that, depending on the perturbation, one KS orbit may in fact represent a number of Keplerian periods owing to the slow KS procedure discussed above. The early phase of this work employed the stability criteria of Harrington (1977) and Eggleton & Kiseleva⁷ (1995), with an empirical fitting formula based on three-body calculations for a limited range of parameters. We have now implemented the general stability criterion of equation (90) which specifies the smallest permitted pericentre for adopting an unperturbed inner binary. This new criterion yields more stringent limits than the earlier criteria, especially for large values of e_{out} .

Provided the condition for hard binding energy is also met, which indicates a robust orbit, a triple is said to be stable if the condition $R_{\text{p}}^{\text{out}} > R_{\text{p}}^{\text{crit}}$ is satisfied, whereupon the inner binary is replaced by the centre of mass and its motion is combined in a KS solution with the outer component. We refer to such a system as a *merged* triple in which the inner binary is *dormant* temporarily, that is, it does not experience any dynamical effect (whereas it does evolve as a result of mass loss or change of stellar radii). However, this may not be the case if the inner binary is close enough to interact tidally, or if the inner and outer orbits are inclined, in which case the inner eccentricity can be induced to higher values while its semimajor axis remains secularly constant (in the absence of damping). These important cases are discussed in the following.

The merger treatment is terminated and the inner binary re-initialized upon one of the following developments: (i) the outer orbit becomes strongly perturbed (i.e. $\gamma > 0.25$), or (ii) the pericentre condition is violated at subsequent *apocentre* passages. Various correction procedures are also carried out at the start and termination in order to maintain formal energy conservation and be consistent with the new equations of motion. Termination resulting from large perturbations does not necessarily destroy a hierarchical configuration; hence many systems may persist over significant times even after several strong fly-by interactions. For this reason a special algorithm is required to identify newly formed systems and some provisional results have been presented (de la Fuente Marcos et al. 1997).

Equation (90) represents an *upper* limit for the stability of non-coplanar systems, that is, systems whose outer and inner orbits are inclined will be more stable (they may be up to around 30 per cent closer at outer periastron) than coplanar prograde systems with the same mass ratios and eccentricities. Thus we employ an additional

⁶ The quantity $E_{\text{in}}/\Omega_{\text{in}}$ is an adiabatic invariant, where E_{in} is the binding energy of the inner binary and Ω_{in} is its orbital frequency.

⁷ A sign error is corrected in Kiseleva et al. (1996).

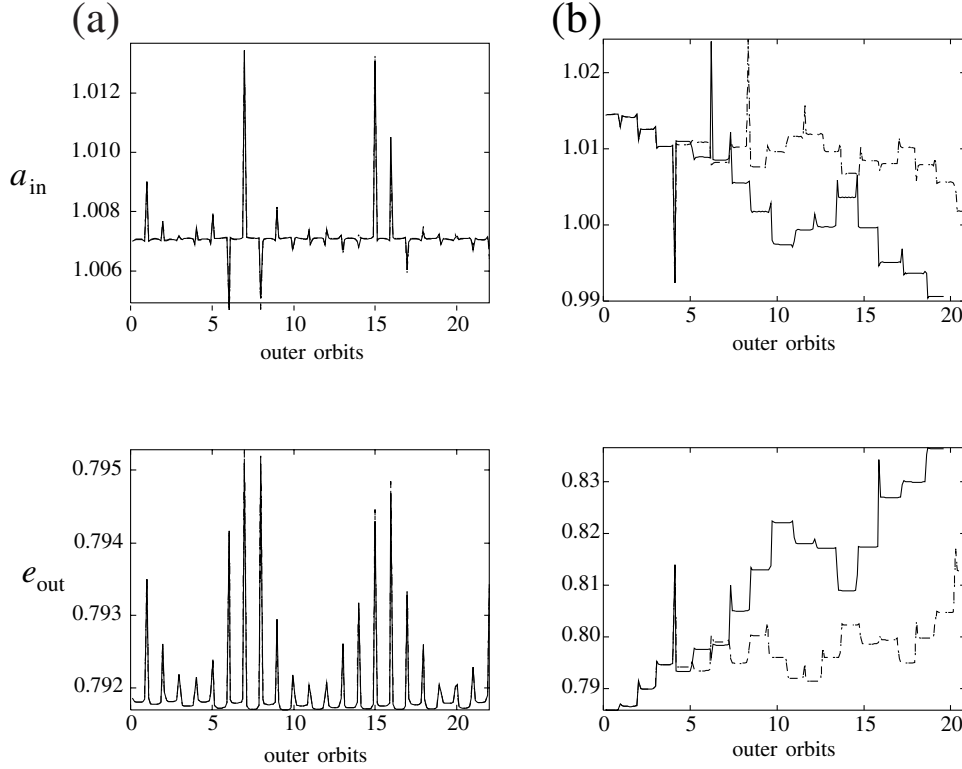


Figure 11. Stability versus instability: non-chaotic and chaotic behaviour. Both compare $e_{in} = 0$ and $e_{in} = 0.00001$ together with $e_{out} = 0.8$ and (a) $R_p^{out} = 5$ and (b) $R_p^{out} = 4$. Despite large fluctuations in the semimajor axis in (a), which reflects energy transfer during outer periastron passage, there is no secular change. In contrast chaotic energy exchange occurs in case (b), this being diagnostic of the inevitable escape of one body.

ad hoc linear reduction factor $f = 1 - 0.3i/180$ (with i in degrees) to account for the increased stability of inclined and retrograde motion. This factor is obtained from numerical experiments (Mardling & Aarseth 1999) and is in good agreement with the increased stability already found by Harrington (1972) for coplanar retrograde orbits. Systems which are still outside the modified criterion may also be accepted, provided at least 10^4 inner apocentre passages have elapsed with R_p^{out} falling inside an interval of *marginal* stability defined by $[0.6R_p^{crit}, fR_p^{crit}]$. However, such cases are extremely rare and this procedure should be considered experimental.

4.2 Higher-order systems

The stability of higher-order systems may also be described by equation (90). For example, the interaction of a binary and a triple may produce a stable quadruple, which will consist of a pair of binaries (2 + 2) or a double hierarchy (3 + 1). In the 2 + 2 case, the more tightly bound binary plays the role of the outer body in a triple, while the triple in the 3 + 1 configuration plays the role of the inner binary, with its own inner binary acting as a single object. In the binary-tides case, it was shown in Mardling (1991) that the chaos boundary is not very sensitive to whether or not the binary is composed of a point mass and an extended object, or two extended objects. Although we have not tested the applicability of the criterion for the stability of quadruples, we expect it to hold approximately, even in the case of a pair of identical binaries. In practice, for 2 + 2 systems we adopt a correction factor $f_1 = f + 0.1 \min(a/a_2, a_2/a)$, where a_2 denotes the second semimajor axis. It is also necessary to ensure that the largest semimajor axis is used in (90), together with the appropriate mass ratio. Hence

such systems are treated in an analogous way to the case of an outer point-mass particle. So far, procedures for stable hierarchies containing up to six members have been implemented.

4.3 The Kozai process

If the outer orbit is inclined to the orbit of the inner pair, it is possible for the inner eccentricity to be induced to a high value, at least in a system of point masses. This effect was noted by Kozai (1962) and is essentially a result of two factors. Firstly, the inner orbit rocks (nutates) in response to the torque applied by the third star, an effect observed in spinning tops torqued by the gravitational field of the Earth. Secondly, the inner semimajor axis is secularly constant in stable triples. These two properties together with conservation of angular momentum lead to a cyclic variation in the inner eccentricity such that the quantity $\cos^2 i (1 - e_{in}^2)$, where i is the relative inclination of the orbits, is conserved in the limiting case in which the outer orbit is not affected by the inner orbit and assuming the absence of external perturbations.

The time-scale for a ‘Kozai cycle’ is given by (Heggie, private communication)

$$T_{\text{Kozai}} = \frac{T_{\text{out}}^2}{T_{\text{in}}} \left(\frac{1 + q_{\text{out}}}{q_{\text{out}}} \right) (1 - e_{\text{out}}^2)^{3/2} g(e_{\text{in}}, \omega_{\text{in}}, i), \quad (91)$$

where T_{in} and T_{out} are the inner and outer orbital periods of the triple, ω_{in} is the argument of periastris of the inner binary and g is of order unity except when $e_{\text{in}} \leq 0.1$ and $i \geq 40^\circ$, in which case it tends to infinity at $e_{\text{in}} = 0$. This result comes from first-order analysis; higher-order analysis produces a finite value for T_{Kozai} when $e_{\text{in}} = 0$. Another constant of the motion in the limiting case

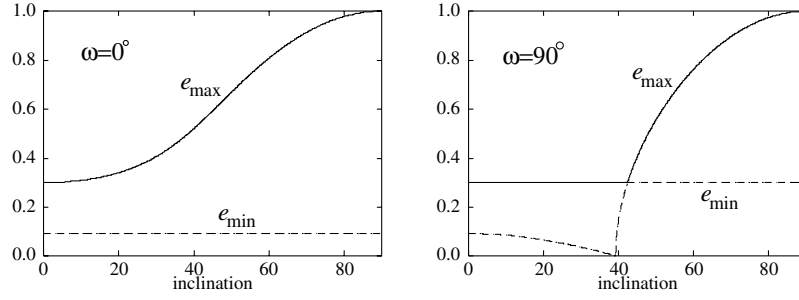


Figure 12. The maximum and minimum values of e_{in} versus initial inclination for initial $e_{\text{in}} = 0.3$ and two extreme values of ω_{in} .

is the quantity $\sin^2 i [5e_{\text{in}}^2 \sin^2 \omega_{\text{in}} + 2(1 - e_{\text{in}}^2)]$. The tidal field of the third body produces apsidal motion in the inner binary such that this quantity is conserved, again provided there are no external perturbations. Fig. 12 shows the maximum and minimum values of e_{in} for an initial value of $e_{\text{in}} = 0.3$ and two different values of ω_{in} . These curves were calculated using formulae supplied by Heggie (private communications).

In order to avoid direct integration of stable hierarchies, we first tried a simple treatment of the Kozai process. If the eccentricity growth time is relatively short (say $< 1 \times 10^7$ yr), we use an analytical expression for $\dot{e}_{\text{in}}$, derived by averaging over the inner and outer orbit, to modify the inner eccentricity (at constant a_{in}) by an amount $\Delta e_{\text{in}} = \dot{e}_{\text{in}} \Delta t$, where the time interval is given by $\Delta t = 0.1(1 - e_{\text{in}})T_{\text{Kozai}}$. This is done until e_{max} or e_{min} is reached, or until $\dot{e}_{\text{in}}$ becomes comparable to $\dot{e}_{\text{circ}}$, that is, $\dot{e}$ given by (18) (see next section). Associated changes to the inclination of the inner orbit are implemented using the first Kozai integral. Differentiation gives $\Delta i = -e_{\text{in}} \Delta e_{\text{in}} / (1 - e_{\text{in}}^2) \tan i$, where the inclination is given by $\cos i = \hat{\mathbf{h}}_{\text{in}} \cdot \hat{\mathbf{h}}_{\text{out}}$ and $\mathbf{h}_{\text{in}} = (\mathbf{r} \times \dot{\mathbf{r}})_{\text{in}}$, and similarly for $\mathbf{h}_{\text{out}}$. Using standard KS transformations, the orbital elements of the dormant inner binary are then modified in a consistent manner by small changes in response to the external torque.

The adopted procedure ensures a gradual adjustment coupled to the average growth time, and relatively long-lived stable configurations may therefore experience several Kozai cycles with large maximum eccentricity. In the case of externally applied corrections for systems not satisfying the stability conditions, no distinction is made between standard binaries and those already undergoing tidal circularization; i.e. it is merely a question of time-scale. The only significant difference is that the latter are re-initialized according to Section 3.4. Thus it appears feasible to model the process of induced eccentricity change of any relevant binary by analytical means in order to maintain a more realistic description of hierarchical systems. Although this scheme should be considered experimental and has actually been replaced (see below), it reflects the desire to explore the adaptation of analytical eccentricity modulations which would otherwise be extremely time-consuming by direct calculations.

4.4 Tidal interactions

If the inner binary is close enough to interact tidally, the associated apsidal motion can ‘detune’ the Kozai effect, that is, e_{in} and i are no longer related via the integrals above and the maximum value for e_{in} is less than that given by the Heggie formula. Detuning is effective if the apsidal motion period resulting from tides is comparable to or less than that owing to the presence of a third body. For systems in which the inner binary is

not nearly circular, the ratio is given by (Schwarzschild 1965; Innanen et al. 1997)

$$\frac{T_{\text{Kozai}}}{T_{\text{tide}}} \sim 15k_1 q_{\text{in}} \left(\frac{T_{\text{out}}}{T_{\text{in}}} \right)^2 \left(\frac{1 + q_{\text{out}}}{q_{\text{out}}} \right) \frac{(1 - e_{\text{out}}^2)^{3/2}}{(1 - e_{\text{in}}^2)^5} \left(\frac{r_1}{a_{\text{in}}} \right)^5 \\ \sim \frac{15k_1}{(1 + e_{\text{in}})^5} \left(\frac{1 + e_{\text{out}}}{1 - e_{\text{out}}} \right)^{3/2} \left(\frac{q_{\text{in}}}{q_{\text{out}}} \right) \left(\frac{R_{\text{p}}^{\text{out}}}{a_{\text{in}}} \right)^3 \left(\frac{r_1}{R_{\text{p}}^{\text{in}}} \right)^5, \quad (92)$$

where R_{p}^{in} is the periastron separation of the inner binary, and we are assuming (just for this discussion) that only star 1 contributes to the apsidal advance resulting from tides. To illustrate with a simple example, for a triple with inner and outer eccentricities of 0.5 and equal masses, if $k_1 = 0.01$ and $R_{\text{p}}^{\text{out}}/a_{\text{in}} = 10$, the time-scales are equal if $R_{\text{p}}^{\text{in}}/r_1 \sim 3$.

It is the quadrupole moment induced in a star by its companion which produces apsidal motion in a binary. The stellar spin also induces a quadrupole moment, and if the spin rate is high, this effect can dominate the apsidal motion. For a spin frequency of Ω , we can define a parameter β such that

$$\Omega^2 = \beta \frac{G(m_1 + m_2)}{R_{\text{p}}^{\text{in}^3}}, \quad (93)$$

that is, the star is in synchronous rotation with the orbit at periastris for $\beta = 1$. The ratio of time-scales for apsidal advance resulting from spin and tides is

$$\frac{T_{\text{spin}}}{T_{\text{tide}}} \sim \frac{15}{\beta} \frac{q_{\text{in}}}{1 + q_{\text{in}}}. \quad (94)$$

Thus for synchronous rotation in an equal-mass system, the tides are slightly more dominant. Therefore (and for reasons of simplicity at this stage), we do not include the effect of stellar spin in our model, although we acknowledge its importance in the tidal evolution process.

The presence of a third star in a bound subsystem introduces considerable complexity to the tidal interaction process. However, we have chosen to include the evolution of such systems in an attempt to model more accurately interactions of astrophysical interest (i.e. orbit shrinking) which would otherwise be ignored. Since such interactions often produce relatively close binaries, the Kozai process may lead to an unrealistic number of stellar collisions if tidal friction is not included. We have used an averaging procedure devised by Heggie (private communication) to calculate the evolution of such systems. The following relatively expensive implementation is only applied to those systems not already undergoing circularization for which $e_{\text{max}} > 0.8$. Thus binaries with a large formal value of e_{max} and short Kozai cycle will eventually damp and reduce the period in

the standard way. Higher-order systems are excluded from the treatment for technical reasons.

At each apocentre of an outer perturbed orbit, or before each check of unperturbed motion, we determine the relevant orbital elements. During the subsequent interval, only the inner binary elements are recalculated since the back effect on the outer orbit is neglected. Given the averaged evolution equations for the Runge–Lenz vector $\mathbf{e}$ and the angular momentum per unit mass vector $\mathbf{h}$, we employ a fourth-order Runge–Kutta integrator to calculate the evolution of these vectors. They in turn provide the orbital elements (except for the phase) and hence the physical coordinates and velocities, where we choose the phase to be such that the system is placed at periastron which may facilitate other procedures. Finally, the saved KS variables $\mathbf{u}$ and $\mathbf{u}'$ are calculated. The basic time-step used for this procedure is given by $\Delta t = \frac{1}{2}(P_{\text{in}} T_{\text{Kozai}})^{1/2}$, with a check on the first Kozai integral for unperturbed *outer* motion showing satisfactory conservation during significant eccentricity changes in the case of no damping. Additional checks on the time-step and pericentre ratio are included in order to avoid an unduly large eccentricity increase near maximum, when the tidal contributions (or apsidal motion) may be significant.

The hierarchical configurations are terminated temporarily if $t_{\text{circ}} < 2 \times 10^9$ yr. Consequently, the standard treatment is initiated upon a successful test for chaotic or regular tidal interaction, whereupon the new hierarchy is accepted again after a possible short delay involving chaotic motion.

It can be seen from the above that hierarchical systems provide a fascinating challenge for numerical modelling. This is a complicated problem which requires careful study. In particular, the difficult question of distinguishing between absolute and long-term stability needs to be addressed. Thus on some occasions a hierarchy which is not accepted as stable may survive for significant time intervals and even maintain a remarkable constancy of the inner semimajor axis. Since direct integration of borderline systems can be very expensive, especially for hard binaries with large values of e_{out} , it is desirable to formulate criteria for *practical* stability that might be applied over more limited intervals. Some experimentation along these lines have already been attempted. However, the unpredictable nature of chaotic onset of instability presents obstacles which must be overcome before such an approach can be justified.

5 APPLICATIONS

By now, a large number of models have been investigated and much experience gained. However, the present attempt at consistent star cluster simulations involves many aspects of stellar evolution, as well as the need to study different processes on a wide range of time-scales which require a variety of algorithms. Consequently, the code implementations present a formidable challenge which will necessitate further refinements. It is fortuitous that the treatment of chaotic and normal tidal evolution does not add significantly to the computational overheads, since with the present binary parameters the number of such cases is relatively small with each one usually requiring a modest numerical effort. In any case, increasing the binary fraction as well as the maximum period will pose a greater strain on the computational resources than the astrophysical overheads.

In the following, we illustrate the role of tidal dissipation by discussing some results from one of the most recent models, specified in Section 3.1. This cluster model has an initial

half-mass radius $R_h \approx 3.3$ pc, total mass $7.4 \times 10^3 M_\odot$ and a lifetime until complete disruption of $\sim 5.9 \times 10^9$ yr. During this time, some 18 chaotic and 55 normal tidal events occurred which form a useful sample for study. The overall evolution is characterized by a small preferential depletion of single stars with respect to binaries, although the latter are diminished by collisions, neutron star formation, strong interactions and escape. Because of mass segregation, the average binary fraction in the core increases gradually from the initial value of 5 per cent to at least ≈ 20 per cent at the cluster half-life (1.9×10^9 yr) and often exceed 40 per cent during late stages.

Employment of the KS formalism provides information which enables the energy production of hard binaries to be determined after subtracting all contributions of a non-dynamical nature. Thus even the small present binary fraction yields a substantial net gain amounting to ≈ 20 per cent at half-life of the initial energy binding the cluster, some of which represents contributions from compact subsystems. Because of the small cross-sections, a considerable fraction of binaries are not affected by the environment and terminate their evolution in isolation, unless escape intervenes. A full analysis of this network of binary interactions and different stages of evolution would represent a massive undertaking; however, here we highlight some relevant aspects connected with the present work.

It is of interest to identify the mechanism which acts as a trigger for the initiation of chaotic motions. For this purpose we examine the 16 independent events produced by the model used for illustration. Strong interactions often leave an imprint, namely the ejection of a perturber having small impact parameter with respect to the relevant binary, where the outer eccentricity provides a measure of the strength. The onset of chaotic motion appears to be strongly correlated with such processes, and in fact 14 of the events reveal a definite association. Of the two remaining cases, one binary was assigned a large initial eccentricity which increased slightly later and the other acquired its large eccentricity in an exchange interaction shortly before. Moreover, very similar results are exhibited by a subsequent model.

In the sample as a whole, 15 of the eccentricities exceed 0.99, with eight values above 0.999. Given the distribution of primordial hard binaries, it is significant that only four of the chaotic events involve components which were members of the same binary initially. As regards the fate of the chaotic binaries, only two resulted in immediate physical collision, whereas all the others crossed the chaos boundary and were not disrupted by external perturbations. However, there were two additional cases of soft binaries where the KS treatment was first terminated temporarily before being accepted again with similar parameters.

The non-chaotic interactions form another interesting data set for analysis. Out of the total number of 55, some 26 binaries attained complete circularization and two circularizing binaries escaped, while another six became dormant (i.e. large t_{circ}) following mass loss (WD formation) and orbital expansion. The two other main causes of incomplete evolution are binary disruption resulting from strong perturbations and rapid expansion during the giant phase. Although quite small values of semimajor axes are involved (≈ 5 au), these events are mostly connected with the mode of origin; i.e. favourable conditions for normal tidal evolution inside hierarchical systems, followed by rapid disruption.

The duration of the normal tidal evolution is of interest because of the relevance for further dynamical interactions. Here the median time-scale for circularization is $\approx 4 \times 10^7$ yr. Circularizing binaries can present significant cross-sections during this phase,

thus contributing to the energy budget of the cluster. The alternative has been to assume instantaneous circularization (Portegies Zwart et al. 1997). We intend to quantify the importance of this effect in a future paper.

Nearly all the normal tidal binaries which terminated in circular orbits experienced Roche-lobe mass transfer subsequently. In all there were 31 such events, which also included a few cases of early termination as a result of rapid expansion.

The formation of hierarchical systems is of considerable interest. One channel which can be identified is because of strong binary–binary interactions treated by chain regularization. Thus if one binary is disrupted, the remaining triple may in fact be stable according to equation (90) and the subsequent behaviour is more appropriately studied by the so-called merger procedure. The strong interaction process is expected to generate triples with high outer eccentricity, as was first reported in early scattering experiments (Mikkola 1983). However, examples with quite small values of e_{out} have also been noted in our simulations. The importance of stable hierarchies has been recognized for some time (Aarseth 1992) and a provisional analysis of the characteristic features has been attempted (de la Fuente Marcos et al. 1997). Higher-order systems are also formed, in which the *outer* orbit is still inside the hard binary limit, and some 10 independent cases were noted.

6 DISCUSSION

In this paper, we model the evolution of relatively massive open star clusters, taking into account most astrophysical processes known to occur in such clusters. In particular we have focused on tidal circularization, including both the familiar smooth long time-scale circularization of binaries resulting from normal tidal interactions, as well as the process of chaotic energy exchange where huge tides may be generated, providing simple prescriptions for both types of tidal interaction. For normal tidal circularization, the prescription allows us to avoid direct integration of the binary orbit (at the expense of neglecting stellar spin), except when such a binary is perturbed by other stars. Chaotic binary evolution proceeds via a mapping which effectively models the energy exchange at periastron, using an impulse approximation. This continues until the binary has lost sufficient energy through damping to become a normal tidally circularizing binary.

The aims of this study are twofold. On the one hand, we now have a model which allows us to study the dynamical effects of a tidally evolving binary population. The evolutionary path a star cluster follows depends critically on the useful binary fraction in the core, useful binaries being those with substantial binding energies, but not so substantial that their energy is not available to the cluster. Thus the distribution of semimajor axes is important, and its rate of change is also likely to affect the way the cluster evolves. A detailed study of this effect has yet to be done. In the model used for illustration in this study, ~ 10 per cent of the primordial binaries undergo tidal circularization, about half of which circularize completely while remaining in the cluster. Of the remaining half, some are disrupted during violent encounters, while others are destroyed during a common envelope phase. The subsequent evolution of binaries which are tidally evolving when they escape the cluster is not recorded.

At the same time, the present model can be used to study the micro-physics of star clusters. For example, violent chaotic tidal interactions are often induced during energetic dynamical events such as the collision of two binaries. Such an event often produces

a triple configuration, be it temporary or stable, and the evolution of its inner binary has an important bearing on the final state of the system. For example, shrinkage of the inner binary in a triple may stabilize the configuration as long as the circularization time-scale is short enough. For the model described, 16 chaos events were recorded, 14 of which were associated with strong dynamical interactions, with only two resulting in immediate collision. Moreover, all but one involved eccentricities exceeding 0.99, and 12 involved non-primordial pairs. Such events are expected to be even more frequent in globular cluster cores and are possibly the precursors of many of the exotic objects observed in such clusters.

We find that about one in ten models produces a tidal capture event. Although neutron stars are not retained in these models, the occurrence of tidal captures gives support to the hypothesis presented in Mardling (1996a) that some low-mass X-ray binaries in the field are formed in open clusters via tidal capture (as well as via higher-order dynamical interactions). Such evolutionary paths avoid the problems associated with the accepted formation route (Webbink & Kalogera 1994), while still making them extremely rare objects. The modelling of neutron star kicks is based on a Maxwellian velocity distribution with a 1D velocity dispersion of 190 km s^{-1} (Hansen & Phinney 1997).

About one in three models produces a stellar collision between a pair of stars whose orbits are not perturbed by any nearby stars. This result is in good agreement with the prediction made in Davies, Benz & Hills (1991) that the cross-section for collision is about twice that for tidal capture. Such ‘clean’ collisions are much rarer than collisions induced within higher-order systems as, for example, hierarchical triples.

Binaries are selected for normal tidal evolution if their circularization time-scales are shorter than 2.0×10^9 yr. However, the time-scales can also change if one or both component stars expand as they evolve, and/or if mass loss occurs via a stellar wind. In addition, binary orbits may be perturbed through stellar encounters, changing their semimajor axes and hence their circularization times. Binaries within higher-order configurations may also evolve tidally, with extreme scenarios possible, such as described at the end of Section 3.4 (tidal evolution of an inner binary of a hierarchy, followed by exchange, followed by tidal evolution of the new inner binary).

As expected, most isolated binaries which undergo tidal circularization begin doing so once one of their components ascends the giant branch to a point where the stellar separation, in units of the giant radius, becomes sufficiently small. Main-sequence stars which are close enough to interact tidally will usually have done so while on the pre-main sequence, although a few may begin as they expand slightly on the main sequence. Other binaries may be induced to interact tidally following dynamical interactions with other stars. For our sample of primordial hard binaries, over 60 per cent suffered normal tidal circularization during the giant phase of one (or both) stars.

The accurate treatment of binary–binary interactions allows for the creation of both stable and unstable triple configurations; likewise these may interact with singles, binaries or other triples (or indeed higher-order configurations). Although strong interactions between binaries tend to lead to the destruction of one, in addition to loss through coalescence and escape, we find that at all times during the lifetime of the cluster, the core contains a significant proportion of binaries which actually increases as a result of the preferential depletion of massive single stars. The rate of destruction of binaries, and of creation and destruction of high-order configurations depends on the local stellar density and

velocity dispersion. As long as there is enough *hierarchical space* for high-order configurations to exist, stable triples tend to form by binary–binary collisions, whereas quadruples mainly become bound by imparting kinetic energy to passing field stars.

This brings us to the following interesting result: seven blue stragglers observed in the model under consideration were created in unstable hierarchies, with only one additional object formed during Roche-lobe mass transfer. However, other models show some evidence of alternative formation channels; in particular gravitational radiation of unevolved stars which have circularized inside hierarchies and to a lesser extent, classical two-body collisions.

To illustrate the capabilities of the code, we describe an episode in a recent cluster model initially composed of 10 000 single stars and 1000 binaries. It occurs at $t \sim 4$ Gyr when the number of stars has reduced to 1300, and involves a subsystem of five stars. We begin with an encounter between a stable triple and a binary; [(1,2),3] and (4,5). Binary (1,2) with period 22 d is executing Kozai cycles and binary (4,5) has a period of 14 000 d. Star 1 (main sequence) and star 4 (white dwarf) collide, making a red supergiant ($1 + 4 \equiv 14$). This results in a marginally stable high-order system composed of $\{[(14 + 2), 5], 3\}$, with a second and *third* outer semimajor axis of 65 and 200 au respectively. Binary ($14 + 2$) experiences a common envelope stage (owing to the increased radius) which produces a circular binary with semimajor axis of 0.24 au (from 0.9 au) and star 14 becomes a white dwarf. The stability of the double hierarchy can now be examined by the triple stability formula, with the smallest binary treated as a single body: here ($14 + 2$) is that binary. Fifteen million years later, binary ($14 + 2$) interacts with another stable triple [(6,7),8], consisting of a circularizing inner binary of period 16 d; this leads to collision of star 2 and 6, which makes a blue straggler ($\equiv 26$). The inner binary of the resulting *stable* triple [(26,14),8] has a semimajor axis of 0.20 au and an eccentricity of 0.68. This binary executes Kozai cycles, with a maximum eccentricity of 0.90. After 200 million years the hierarchy becomes unstable (following an external perturbation), causing the inner binary to shrink to a semimajor axis of 0.15 au with eccentricity 0.87, followed by ejection of the third star. Its trajectory relative to the binary has an eccentricity of 1.48, indicating an extremely strong encounter (outer periastron distance $2.2a_{\text{in}}$). The interaction activates tidal circularization with significant orbital shrinkage up to the moment of escape. Finally, both the binary (total mass $3.3 m_{\odot}$, $a = 0.03$ au, $e = 0.16$) and the single star (mass $1.0 m_{\odot}$) escape from the cluster (i.e. outside twice the tidal radius) with terminal velocities 20.6 and 65.7 km s^{-1} respectively.

Our direct approach of modelling the whole cluster and the processes therein contrasts with that of Portegies Zwart et al. (1997), who consider the interaction of a binary population with a constant background of single stars representing the conditions in globular cluster cores. Here, encounters between the binaries and single stars are studied by a Monte Carlo selection procedure, together with a wide range of astrophysical outcomes based on a pre-calculated network. However, this preliminary investigation does not include a consistent dynamical evolution of the cluster and also neglects binary–binary collisions which we have found to play an important role in the formation of hierarchical systems. The process of tidal circularization is modelled by instantaneously evolving the semimajor axis at constant angular momentum to the point where the periastron separation is no more than four stellar radii (which may be reached for non-zero eccentricity). On the other hand, our prescription allows for a significantly larger

fraction of binaries to evolve tidally, as well as for such binaries to respond to other internal and external influences as described above. For these reasons, even a general comparison of results is premature and must await further developments. However, we note their general conclusion that binaries evolving in a dynamical environment – albeit a non-consistent one – show a rich diversity compared to an idealized population synthesis.

Although many aspects of star cluster dynamics may be studied using the model described in this paper, some features require further attention. One of the main uncertainties is concerned with the semimajor axis distribution as well as the corresponding mass ratios. To some extent, these parameters may be constrained by detailed modelling of actual clusters, using recent observations, but parameter space still allows many possibilities. The semi-analytical treatment of the Kozai process also needs to be improved in the light of the present experience which is encouraging. Having stressed the importance of hierarchical systems, the question of a primordial population of such configurations arises, and the generation of stable triples has in fact been implemented in the code. As has already been noted, interactions in compact subsystems may produce high-velocity escapers of single stars as well as binaries. It can also be anticipated that such events will be even more pronounced by increasing the proportion of primordial binaries, in accordance with recent observations.

In conclusion, the combination of N -body dynamics with stellar evolution and binary evolution leads to an extraordinary diverse outcome. A full treatment is now essentially in place and together with rapid hardware advances will soon permit direct simulation of small globular clusters. Meanwhile, the present paper has shown that there is also much to be learnt from studying open star clusters which form such fascinating astrophysical laboratories.

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REFERENCES

- Aarseth S. J., 1992, in Duquennoy A., Mayor M., eds, *Binaries as Tracers of Stellar Formation*. Cambridge Univ. Press, Cambridge, p. 6
- Aarseth S. J., 1994, in Contopoulos G., Spyrou N. K., Vlahos L., eds, *Galactic Dynamics and N-Body Simulations*. Springer-Verlag, Berlin, p. 277
- Aarseth S. J., 1996, in Hut P., Makino J., eds, *IAU Symp. 174, Dynamical Evolution of Star Clusters*. Kluwer, Dordrecht, p. 161
- Aarseth S. J., Mardling R. A., 1997, in Johnston S., Walker M. A., Bailes M., eds, *IAU Colloq. 160, Pulsars: Problems and Progress*, ASP Conf. Ser., Vol. 105. Astron. Soc. Pac., San Francisco, p. 541
- Brandt N., Podsiadlowski P., 1995, *MNRAS*, 274, 461
- Davies M. B., Benz W., 1995, *MNRAS*, 276, 876
- Davies M. B., Benz W., Hills J. G., 1991, *ApJ*, 381, 449
- de la Fuente Marcos R., Aarseth S. J., Kiseleva L. G., Eggleton P. P., 1997, in Docobo J. A., Elise A., McAlister H., eds, *Visual Double Stars: Formation, Dynamics and Evolutionary Tracks*, ASSL Ser., Vol. 105. Kluwer, Dordrecht, p. 165
- Eggleton P. P., 1983, *ApJ*, 268, 368
- Eggleton P. P., Kiseleva L. K., 1995, *ApJ*, 455, 640
- Eggleton P. P., Fitchett M. J., Tout C. A., 1989, *ApJ*, 347, 998
- Eggleton P. P., Kiseleva L. G., Hut P., 1998, *ApJ*, 499, 853
- Gingold R. A., Monaghan J. J., 1980, *MNRAS*, 191, 897

- Goldreich P., Soter S., 1966, *Icarus*, 5, 375
Hansen B. M. S., Phinney E. S., 1997, *MNRAS*, 291, 569
Harrington R. S., 1972, *Celest. Mech.*, 6, 322
Harrington R. S., 1977, *AJ*, 82, 753
Heggie D. C., Aarseth S. J., 1992, *MNRAS*, 257, 513
Heggie D. C., Ramamani N., 1995, *MNRAS*, 272, 317
Hut P., 1981, *A&A*, 99, 126
Innanen K. A., Zheng J. Q., Mikkola S., Valtonen M. J., 1997, *AJ*, 113, 1915
Kiseleva L., Aarseth S. J., Eggleton P., de la Fuente Marcos R., 1996, in Milone E. F., Mermilliod J.-C., eds, *The Origins, Evolution, and Destinies of Binary Stars in Clusters*, ASP Conf. Ser., Vol. 90. Astron. Soc. Pac., San Francisco, p. 433
Kochanek C. S., 1992, *ApJ*, 385, 604
Kozai Y., 1962, *AJ*, 67, 591
Kroupa P., 1995, *MNRAS*, 277, 1491
Kroupa P., Tout C. A., Gilmore G., 1993, *MNRAS*, 262, 545
Kumar P., Goodman J., 1996, *ApJ*, 466, 946
Kumar P., Ao C. O., Quataert E. J., 1995, *ApJ*, 449, 294
Kustaanheimo P., Stiefel E., 1965, *J. Reine Angew. Math.*, 218, 204
Lai D., Rasio F. A., Shapiro S. L., 1994, *ApJ*, 423, 344
Lee H. M., Ostriker J. P., 1986, *ApJ*, 310, 176
McMillan S. L. W., Taam R. E., McDermott P. N., 1990a, *ApJ*, 354, 190
McMillan S. L. W., Hut P., Makino J., 1990b, *ApJ*, 362, 522
McMillan S. L. W., Hut P., Makino J., 1991, *ApJ*, 372, 111
Mardling R. A., 1991, PhD thesis, Monash Univ.
Mardling R. A., 1995a, *ApJ*, 450, 722
Mardling R. A., 1995b, *ApJ*, 450, 732
Mardling R. A., 1996a, in Hut P., Makino J., eds, *IAU Symp.* 174, *Dynamical Evolution of Star Clusters*, ASP Conf. Ser., Vol. 90. Kluwer, Dordrecht, p. 273
Mardling R. A., 1996b, in Wijers R. A. M. J., Davies M. B., Tout C. A., eds, *NATO ASI, Evolutionary Processes in Binary Stars*, ASP Conf. Ser., Vol. 90. Kluwer, Dordrecht, p. 81
Mardling R. A., Aarseth S. J., 1999, in Steves B. A., Roy A. E., eds, *NATO ASI, The Dynamics of Small Bodies in the Solar System*, ASP Conf. Ser., Vol. 90. Kluwer, Dordrecht, p. 385
Mikkola S., 1983, *MNRAS*, 207, 115
Mikkola S., Aarseth S. J., 1993, *Celest. Mech. Dyn. Astron.*, 57, 439
Mikkola S., Aarseth S. J., 1996, *Celest. Mech. Dyn. Astron.*, 64, 197
Mikkola S., Aarseth S. J., 1998, *New Astron.*, 3, 309
Podsiadlowski P., 1996, *MNRAS*, 279, 1104
Portegies Zwart S. F., Hut P., McMillan S. L. W., Verbunt F., 1997, *A&A*, 328, 143
Press W. H., Teukolsky S. A., 1977, *ApJ*, 213, 183
Schwarzschild M., 1965, *Structure and Evolution of the Stars*. Princeton Univ. Press, Princeton
Tout C. A., Aarseth S. J., Pols O. R., Eggleton P. P., 1997, *MNRAS*, 291, 732
Verbunt F., Phinney E. S., 1995, *A&A*, 296, 709
Verbunt F., Mardling R. A., Frost C., 1997, unpublished
Webbink R., Kalogera V., 1994, in Holt S. S., Day C. S., eds, *Evolution of X-Ray Binaries*, ASP Conf. Ser., Vol. 90. Am. Inst. Phys., New York
Zahn J. P., 1975, *A&A*, 41, 329
Zahn J. P., 1989, *A&A*, 220, 112
Zare K., 1977, *Celest. Mech.*, 16, 35

APPENDIX A: GENERALIZED PRESS & TEUKOLSKY ANALYSIS

Press & Teukolsky (1977) calculate the energy transfer to the tides of a polytrope in a parabolic orbit with a point mass. This involves a similarity variable, a dimensionless parameter which is constructed from the ratio of the ‘time of periastron passage’ to the dynamical time-scale of the star. Energy transfer will be greatest

when these time-scales are similar. This parameter, which Press & Teukolsky call η , depends on the mass ratio of the perturbing star to the tidally active star, q , and the ratio of the periastron separation to the stellar radius, p :

$$\eta = \left(\frac{p^3}{1+q} \right)^{1/2}. \quad (\text{A1})$$

The energy transfer to the l th mode, δE_l , is given in terms of η by

$$\delta E_l = \left(\frac{Gm_*^2}{r_*} \right) \frac{q^2}{p^{2l+2}} T_l(\eta), \quad (\text{A2})$$

where $T_l(\eta)$ is a dimensionless function which depends on the structure of the polytrope via its natural frequencies of oscillation and overlap integrals (see Section 2), as well as the orbit via η , and m_* and r_* are the mass and radius of the tidally active star.

The time of periastron passage is essentially the reciprocal of the angular velocity at periastron. Using this definition, we can construct (for the moment at least) a new dimensionless parameter, ζ , which depends also on the eccentricity:

$$\zeta \equiv \left(\frac{p^3}{1+q} \right)^{1/2} \left(\frac{2}{1+e} \right)^{1/2} = \eta \left(\frac{2}{1+e} \right)^{1/2}, \quad (\text{A3})$$

where we have introduced a factor of $\sqrt{2}$ so that $\zeta = \eta$ for parabolic orbits. This can be tested by replacing η by ζ in equation (A2) and comparing the energy transfer predicted by this expression with the actual energy transfer to a quiescent polytrope from apastron to the following apastron using the full binary code described in Mardling (1995a). The prescription works very well for eccentricities greater than about 0.9, however, this accuracy can be extended to a wider range of eccentricities via a slightly modified definition of ζ :

$$\zeta = \eta \left(\frac{2}{1+e} \right)^{\alpha(\eta)/2}, \quad (\text{A4})$$

where

$$\alpha(\eta) = 1 + \frac{1}{2} \left| \frac{\eta - 2}{2} \right|^{1.5}. \quad (\text{A5})$$

We can check the accuracy of this approximate similarity variable by choosing various sets of values of p , e and q which leave ζ constant, calculating $T_l(\zeta)$ using the full binary code (and equation A2), then comparing the results with those for $e = 1$ (the ‘correct’ result). Table A1 shows the results of such a test for $\zeta = 3$ for a polytrope of index 1.5. The rows shown in boldface correspond to the correct results. The columns marked ‘per cent’ are relative percentage errors in $T_2(\zeta)$ and $T_3(\zeta)$. The accuracy is good, even for an eccentricity of 0.5 and also for hyperbolic orbits (which are relevant for tidal capture). The accuracy improves as the eccentricity approaches unity. The value for ζ in the row labelled **PT** was obtained using the Press & Teukolsky formula. The small percentage error is a result of the approximation made in their formulation where the orbit remains unchanged during the interaction. Also shown are results for two wider orbits for which $\zeta = 5.3$, the set with $e = 0.9$ corresponding approximately to a point on the chaos boundary. The errors are larger in this case, as tends to be true for wider orbits with fixed eccentricity. But the wider a chaotic orbit is, the closer its eccentricity is to unity, ensuring that the error is never very large.

Table A1. A test of the generalized Press & Teukolsky similarity variable.

ζ	q	p	e	$T_2(\zeta)$	per cent	$T_3(\zeta)$	per cent
3.0	1	2.36	0.5	0.504	4	0.283	8
3.0	1	2.47	0.7	0.495	2	0.275	5
3.0	1	2.57	0.9	0.488	1	0.266	2
3.0	1	2.62	1.0	0.484	–	0.261	–
PT	1	2.62	1.0	0.498	3	0.267	2
3.0	1	2.67	1.1	0.477	1	0.256	2
3.0	1	2.72	1.2	0.470	3	0.250	4
3.0	2	2.82	0.7	0.495	2	0.275	5
3.0	0.5	2.24	0.7	0.492	2	0.274	5
5.3	1	3.70	0.9	0.0187	10	0.0048	15
5.3	1	3.82	1.0	0.0170	–	0.0042	–

APPENDIX B: TABLES OF CURVE FITTING PARAMETERS

Table B1. Quadratic rational function $\tilde{I}(e)$ (equation 22).

i	a_i	b_i	c_i
1	6.306505	32.17211	5.101417
2	–7.297806	13.01598	24.71539
3			–9.627739
4			1.733964
5			–2.314374
6			–4.127795

Table B2. Curve fitting coefficients for equation (46).

c_0	c_1	c_2	c_3	c_4	c_5
-1.763×10^{-3}	0.1277	-3.743×10^{-3}	3.525×10^{-3}	-6.179×10^{-4}	2.744×10^{-5}

Table B3. Maximum periastron separation for tidal capture (Section 2.5).

n	a_0	a_1	a_2	a_3	β
1.5	–600.2227	3538.748	1703.075	77.31540	0.54
3	191.8260	–2.203482	1202.116	70.80281	1.08

Table B4. Coefficients for evaluation of effective polytropic quantities for giants.

i	$a_{i,2}$	$a_{i,3}$	b_i	c_i
0		1.452104	1.934977	–0.3789
1	3.923872	2.214222	–0.392030	1.481
2	–11.88722	–4.855796	6.01655D–02	–0.1018
3	13.46106	4.025394	–3.34790D–03	

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